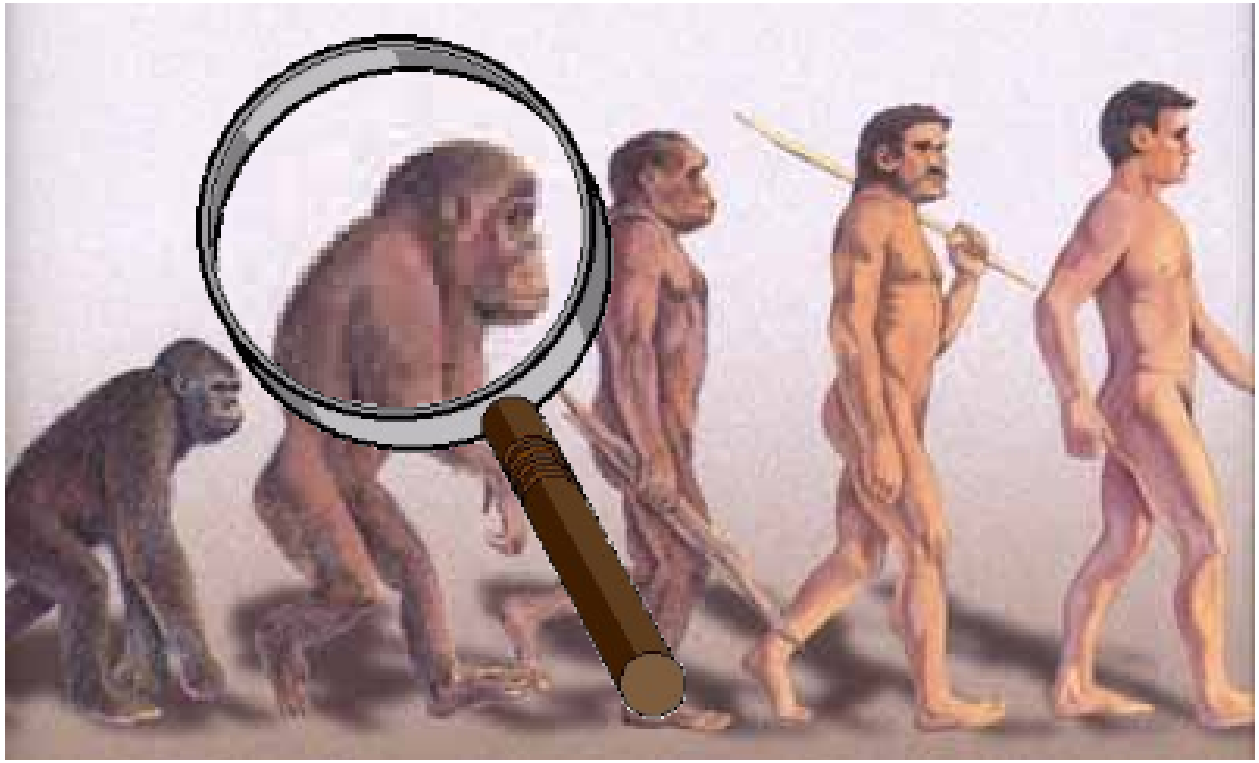
Computational Biology

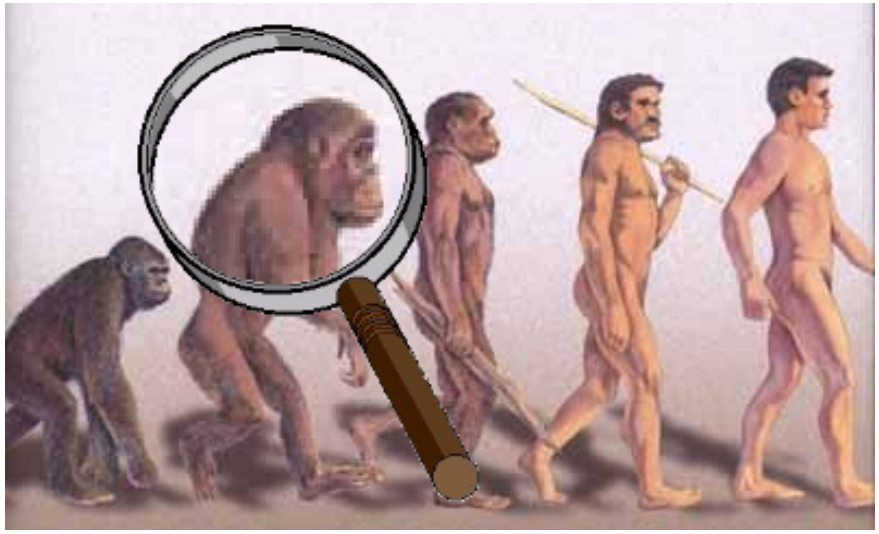
Lecture 11: *Population Genetics*



Fisher-Wright model, ODEs, Markov chains.

Outline

1. **Population Genetics**
2. Facts of Life
3. Infinite Population Model
4. Finite Population Model



Evolution and Molecular Biology

- Evolution is the theory that underpins our understanding of biology
- Molecular biology has open up a whole new perspective on evolution as we see similar proteins in related species
- This evolution ancestry helps us understand molecular biology as similar proteins usually share the same function
- However we can also use molecular biology to help understand evolution—more of this when we discuss phylogenetic tree
- But how does evolution work?

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Evolutionary Population Dynamics

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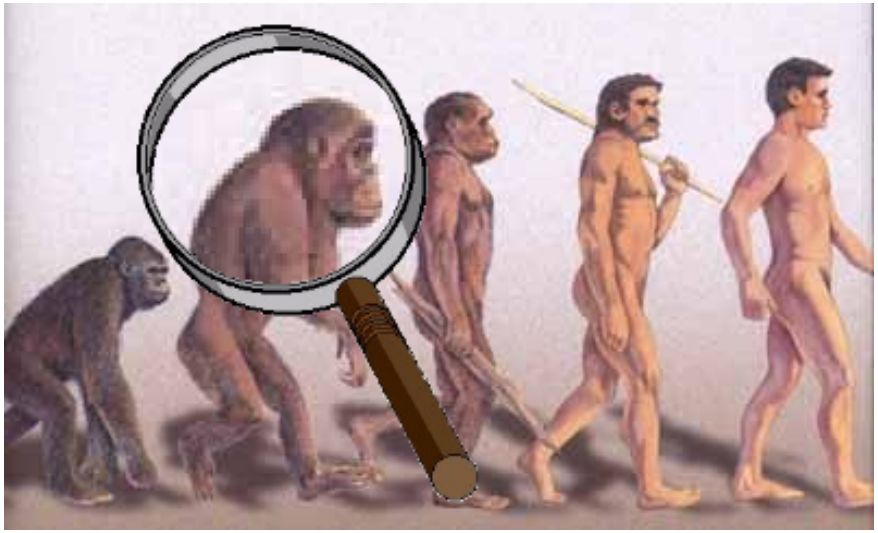

Fisher-Wright Model

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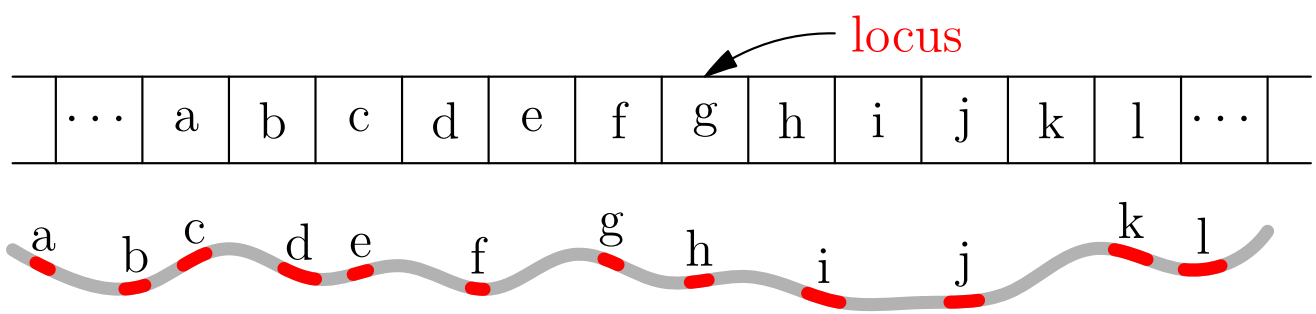
- It is important because it laid the theoretical framework for many of the prejudices held today

- In its simplest form it is a model of what happens at a single locus (site) under selection, mutation and crossover



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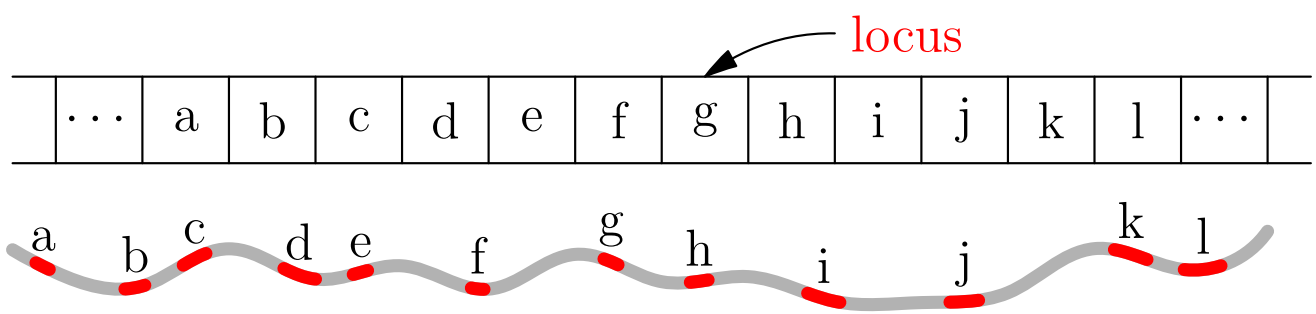

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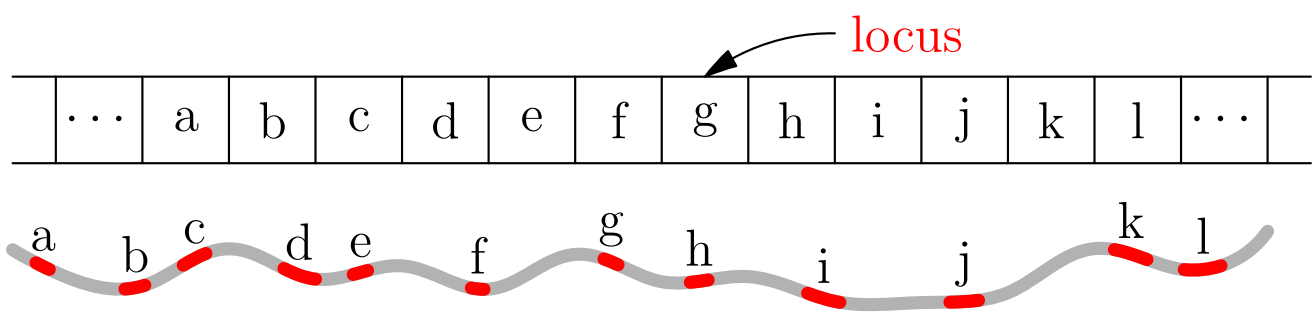

Locus and Alleles

- The locus can be thought of as the location of a gene
- The type of gene at a site will differ between individuals (e.g. one for smooth peas and one for wrinkled peas)
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- We assume that genes can be modified from one allele to another by mutation
- Alleles are also exchanged during sexual reproduction (crossover)

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- If one individual is 10% fitter than another it will (in expectation) produce 10% more offspring
- Of course, the individual may get flattened by a car before it reaches sexual maturity and never produce any offspring
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- Thus the fitness of an allele will typically depend on other alleles at other locations
- This interaction is known as **epistasis**
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Sex

- Sex is surprisingly ubiquitous in higher organisms
- Few higher organisms seem to reproduce asexually (although even if they do, many sometimes also reproduce sexually)
- Asexual higher organisms all would have produced sexually sometime in their past
- Why sex is surprising is that if females produce asexually (producing clones) then they breed at twice the rate of mixed couples
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Linkage Equilibrium

- As well as no epistasis the model assumes lots of crossover (compared to selection and mutation)
- This is not unrealistic in many cases
- Under this assumption we can treat the loci independently and concentrate on a single locus
- This is known as **linkage equilibrium**
- We therefore only need to consider the proportion of the different alleles at the locus
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- Like sex, it is not altogether clear what advantage diploidy brings
- Clearly, diploids can tolerate mutations that make one set of genes dysfunctional
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- Organisms that do this are said to be **haploidy**
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Gender

- Life tends also to be gendered
- This possibly just reflects that finding a mate consumes energy and by specialisation some energy can be saved
- However, this is another complicating factor, which adds little to the model
- I will assume a **monoecious** population, i.e. one with only one gender, although it still mates
- These are all simplifying assumptions, but it allows us to concentrate on the role of selection, mutation and crossover

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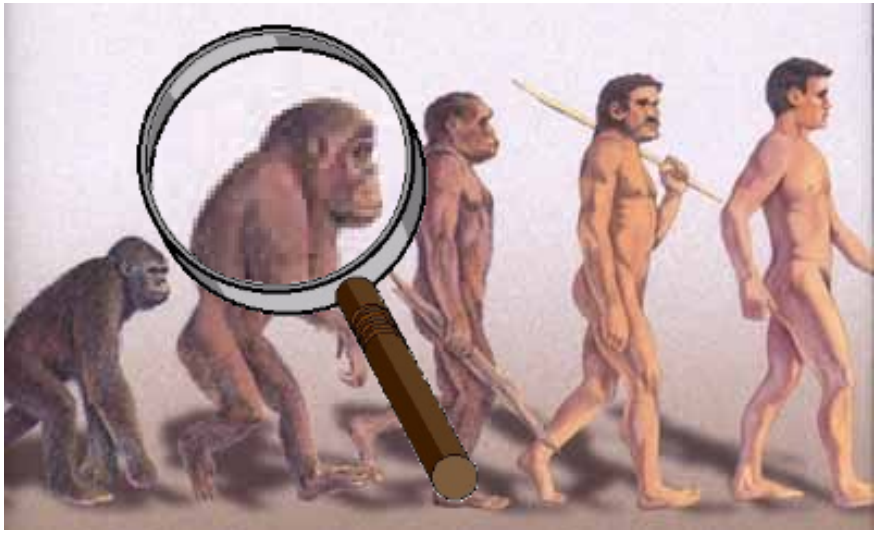

Single Locus Model

- We consider the case where the allele can take just two states A and B
- We call B the mutant
- We make the following assumptions
  - ⋆ B has a selective advantage of $1 + s$
  - ⋆ A mutates to B with a forward mutation rate of u
  - ⋆ B mutates to A with a backward mutation rate of v
  - ⋆ The population size, P, is unchanged from generation to generation

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Many Seeds

- We consider the population changing from one generation to the next by a three stage process
  - ⋆ Each individual produces a very large number of seeds in proportional to its fitness (selective advantage)
  - ⋆ The seeds are mutated according to the mutation probabilities
  - ⋆ We select a sample of P of the seeds independently to form the next generation
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Dynamics

- Assume we start with n mutants
- After selection we have a proportion p_s of mutant seeds

$$p_s = \frac{(1+s)n}{(1+s)n + P - n} = \frac{(1+s)n}{P + s\,n}$$

- After mutation we have a proportion p_{sm} of mutants

$$p_{sm} = (1-v)p_s + u(1-p_s) = \frac{(1-v)(1+s)n + u(P-n)}{P + s\,n}$$

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- If we assume an infinite population then this will be the proportion of individuals at the next generation

$$p(t+1) = \frac{(1-v)(1+s)p(t) + u\big(1-p(t)\big)}{1+s\,p(t)}$$

- If the effect of selection and mutation is small then we can approximate the dynamics by a differential equation

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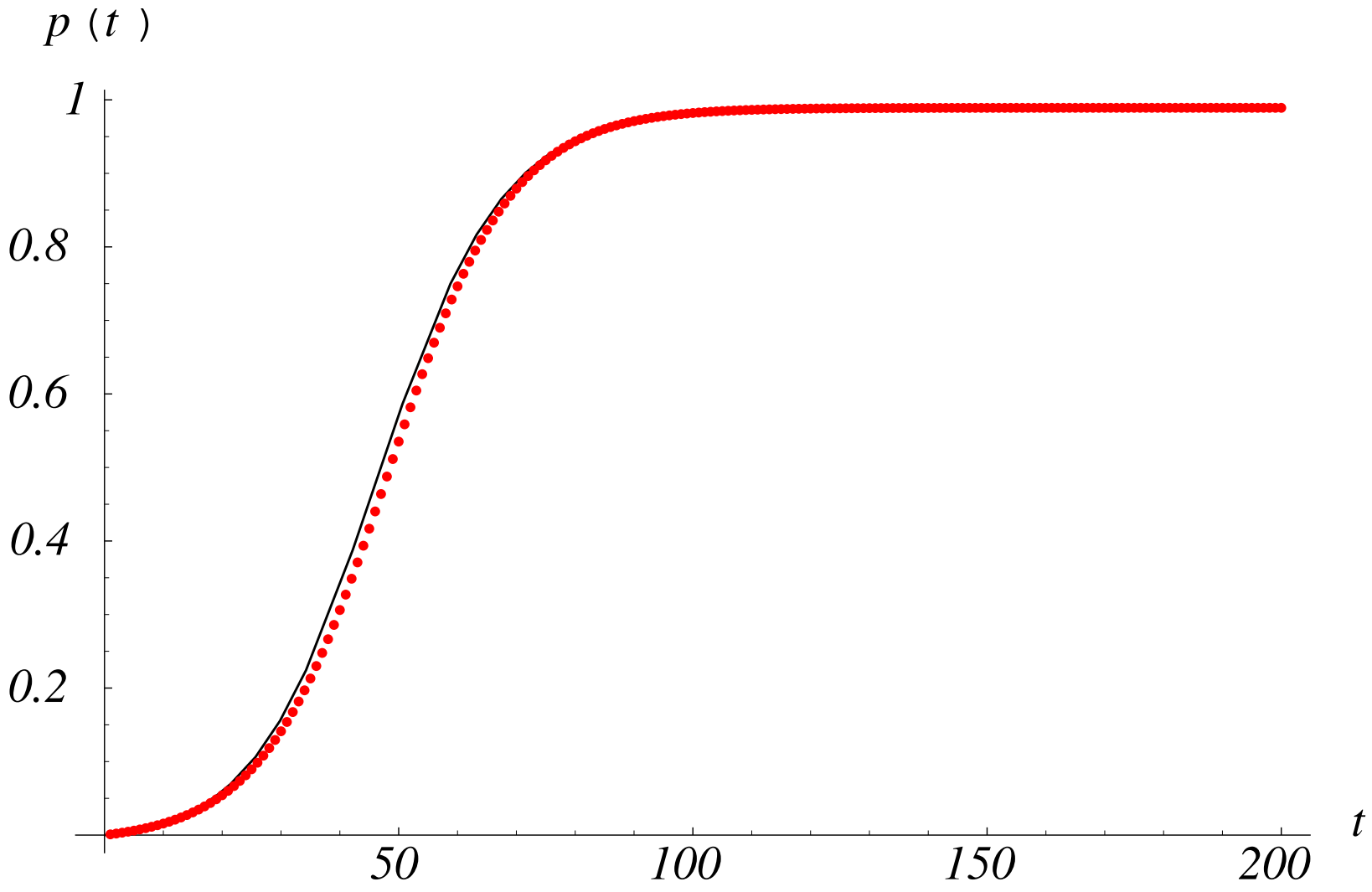
Infinite Population Prediction



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Difficulties of Infinite Populations

- In an infinite population a mutation (even if it is very unlikely) will always occur in a small fraction of individuals
- Selection can rapidly amplify the mutants if they are fitter—note fitter individuals initial grow exponentially

$$(1+s)^t = \mathrm{e}^{t\,\log(1+s)} \approx \mathrm{e}^{s\,t}$$

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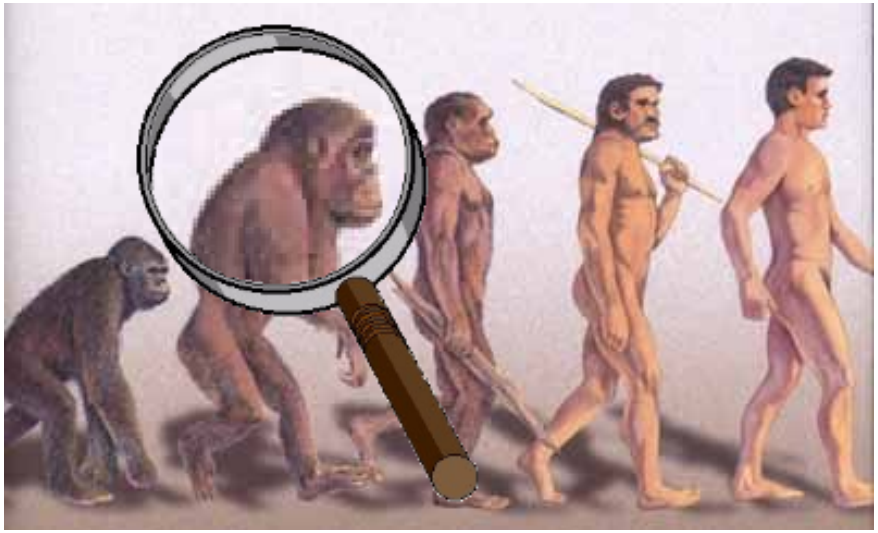
Outline

1. Population Genetics
2. Facts of Life
3. Infinite Population Model
4. **Finite Population Model**



Finite Populations

- To model finite populations, we have to draw a sample of P individuals from the seeds

- The probability of drawing $n' = n(t+1)$ mutants given a proportion $p_{sm}(n(t))$ of mutants in the seed population is given by a binomial distribution

$$\mathbb{P}\left(n'\right) = \binom{P}{n'} p_{sm}^{n'} \left(1 - p_{sm}\right)^{P-n'}$$

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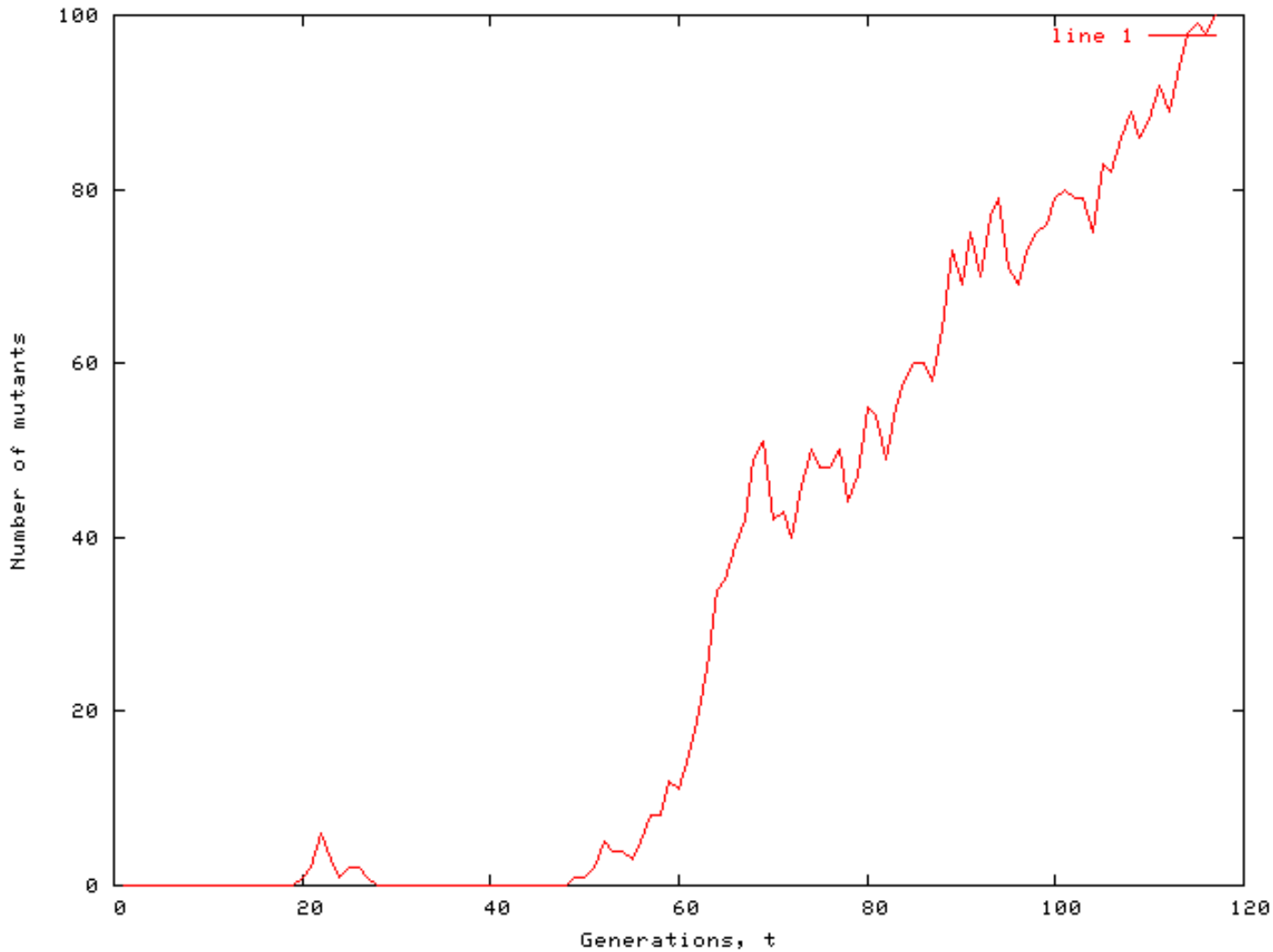
Finite Population Run



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Markov Models

- The evolving population can be modelled as a Markov chain
- The population at time t depends only on the population at time $t-1$
- It is *conditionally independent* of the population at earlier times *given* the population at time $t-1$
- We define $p_n(t) = \mathbb{P}\left(n(t) = n\right)$ to be the probability that there are n mutants at time t
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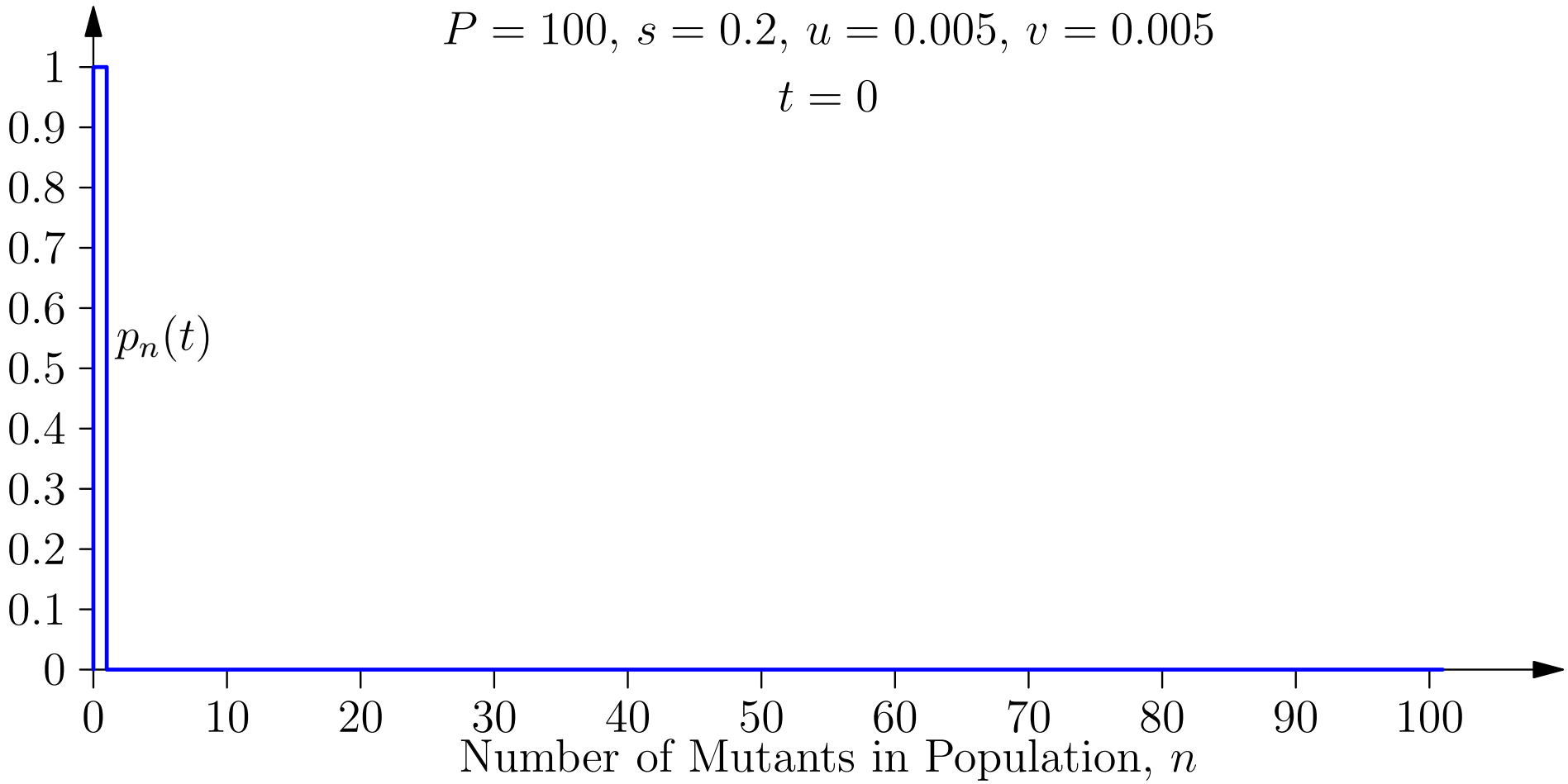
Example of Markov Analysis



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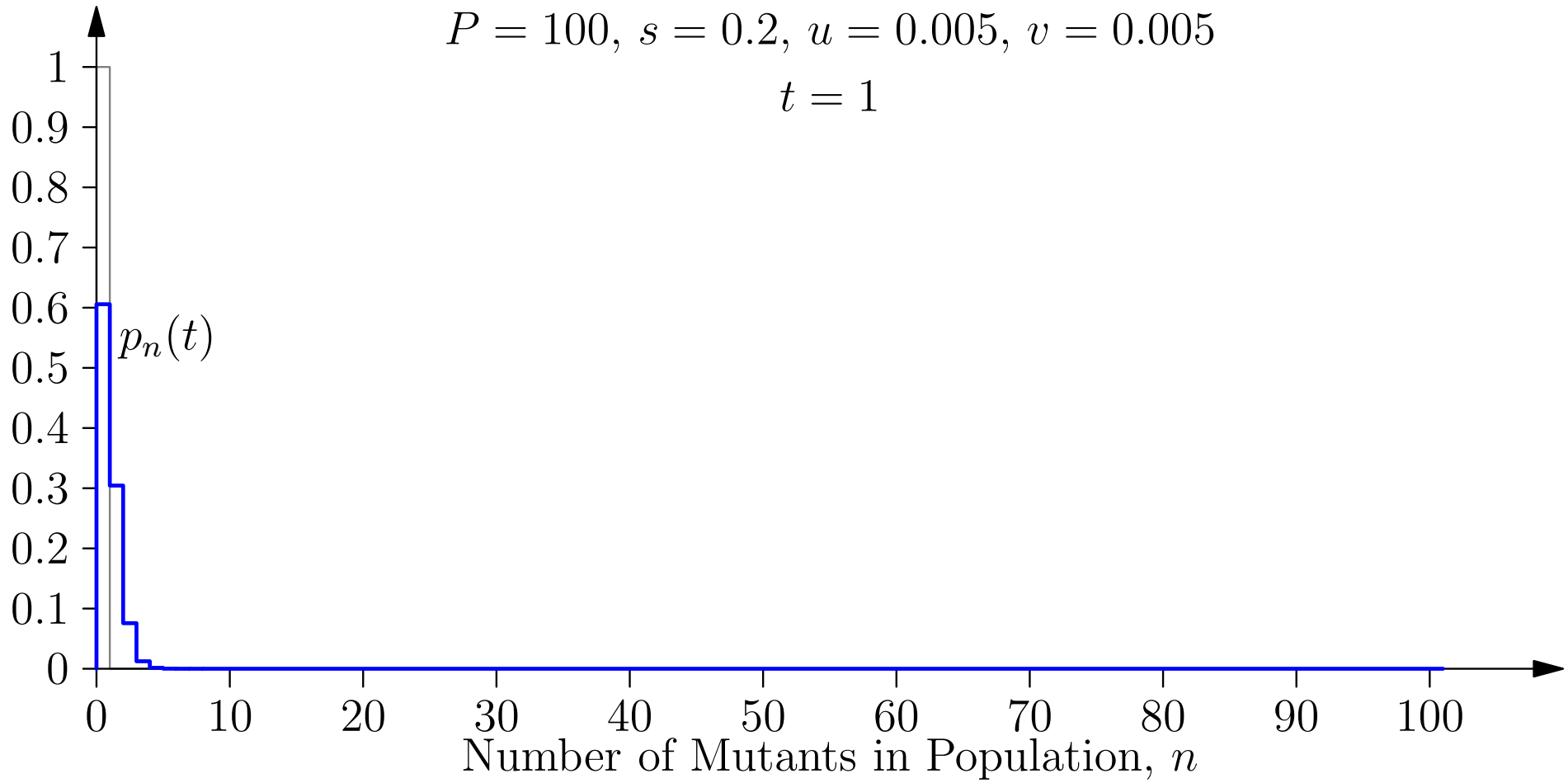

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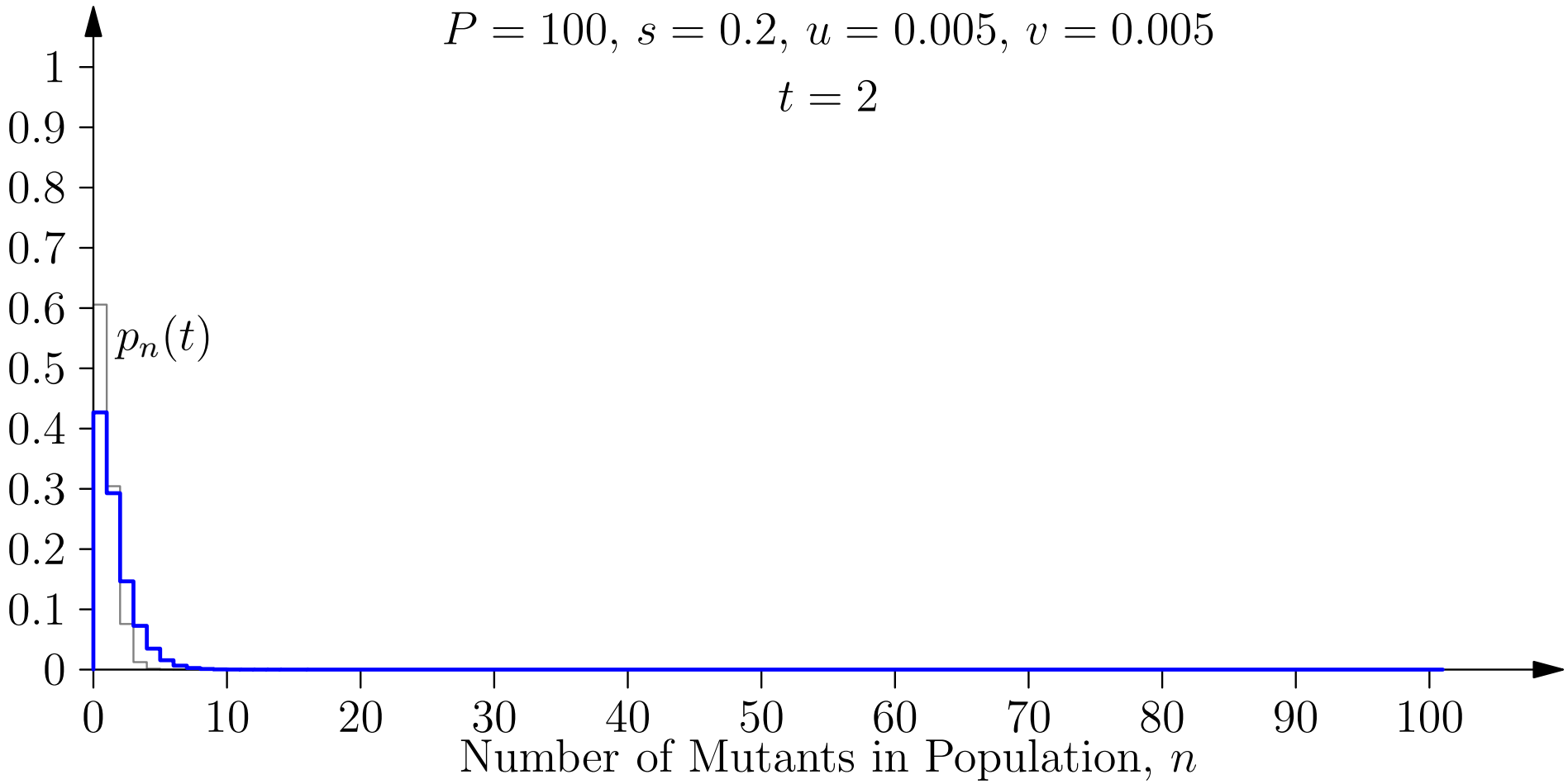

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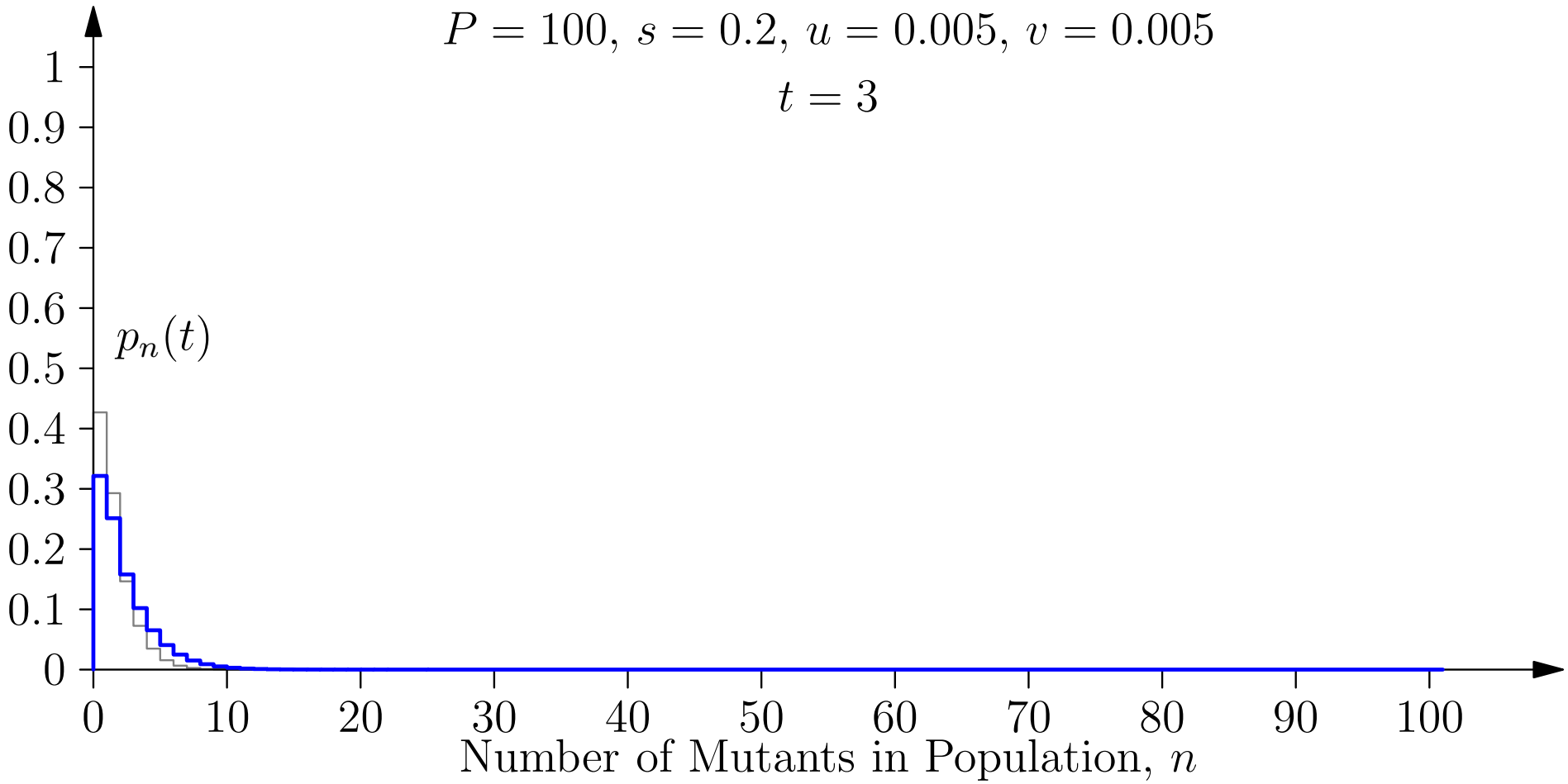

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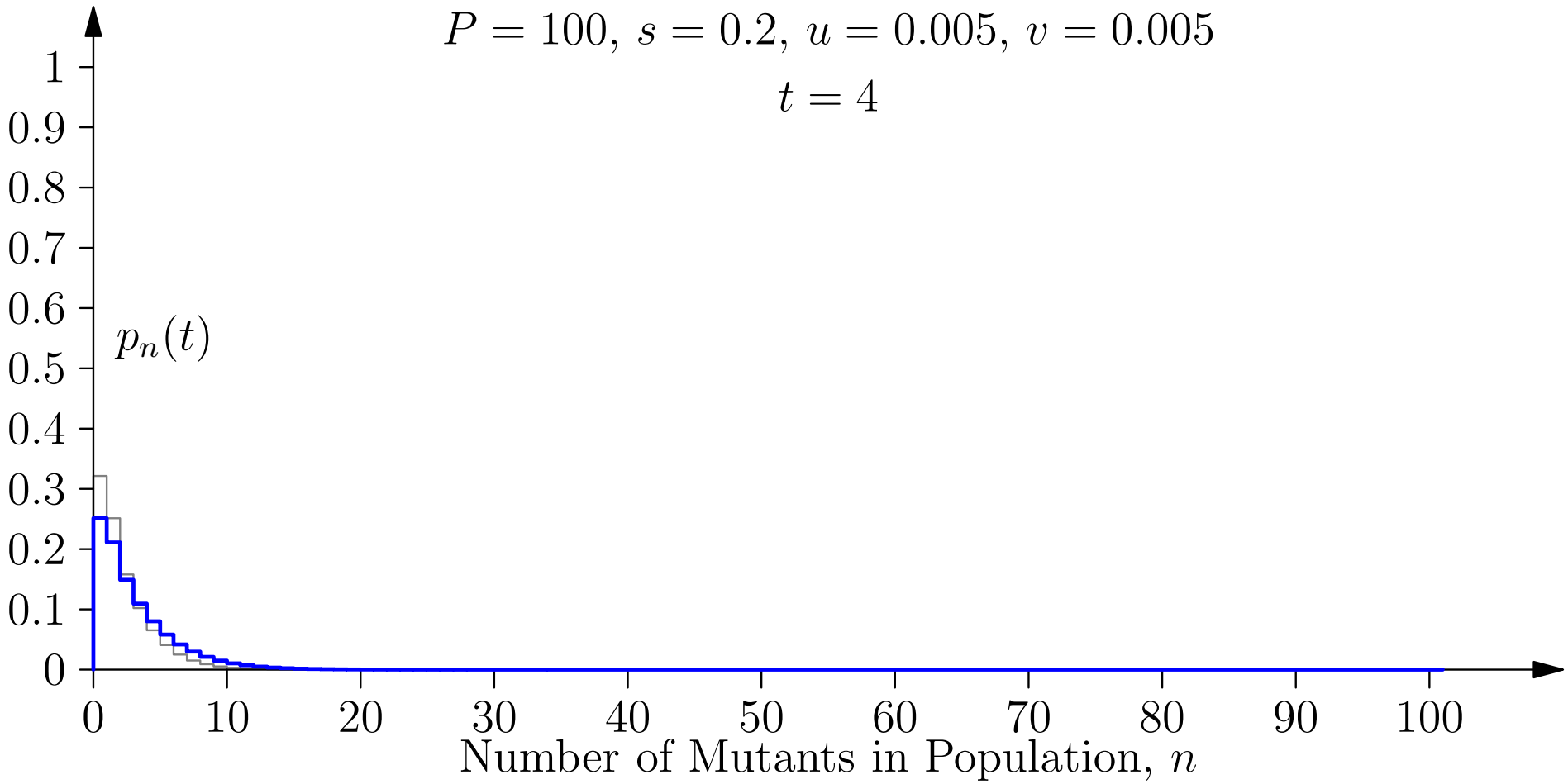

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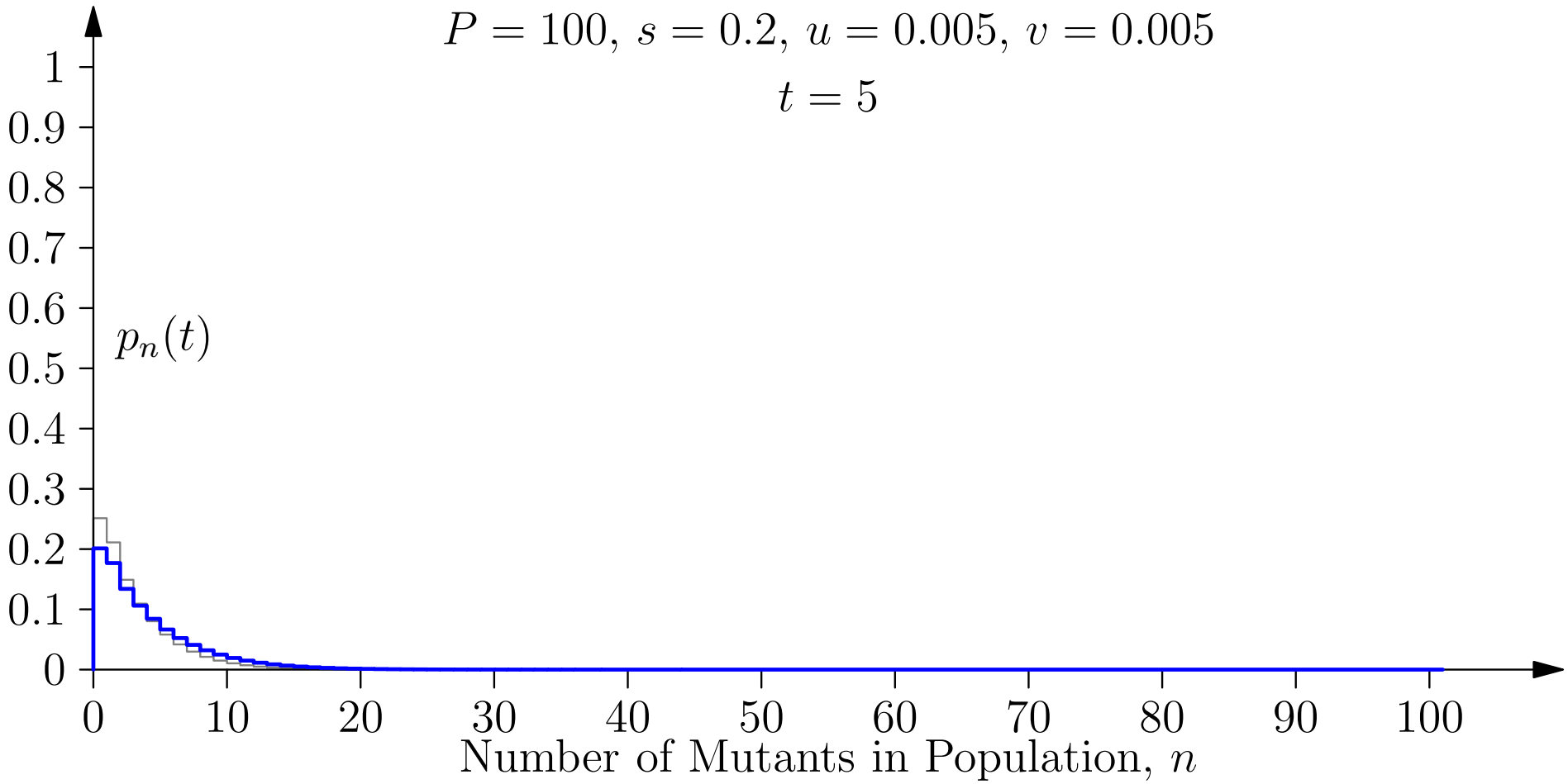

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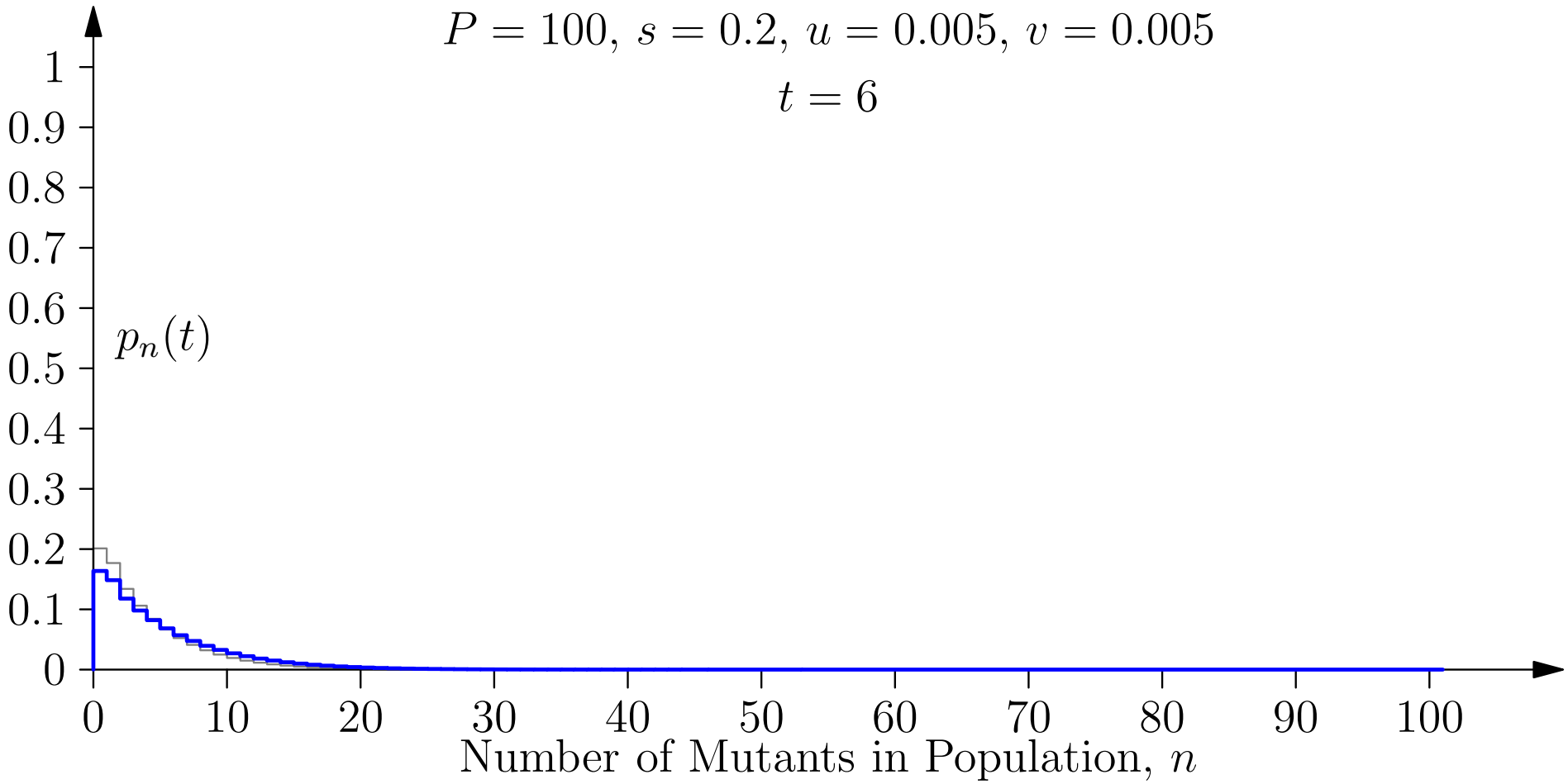

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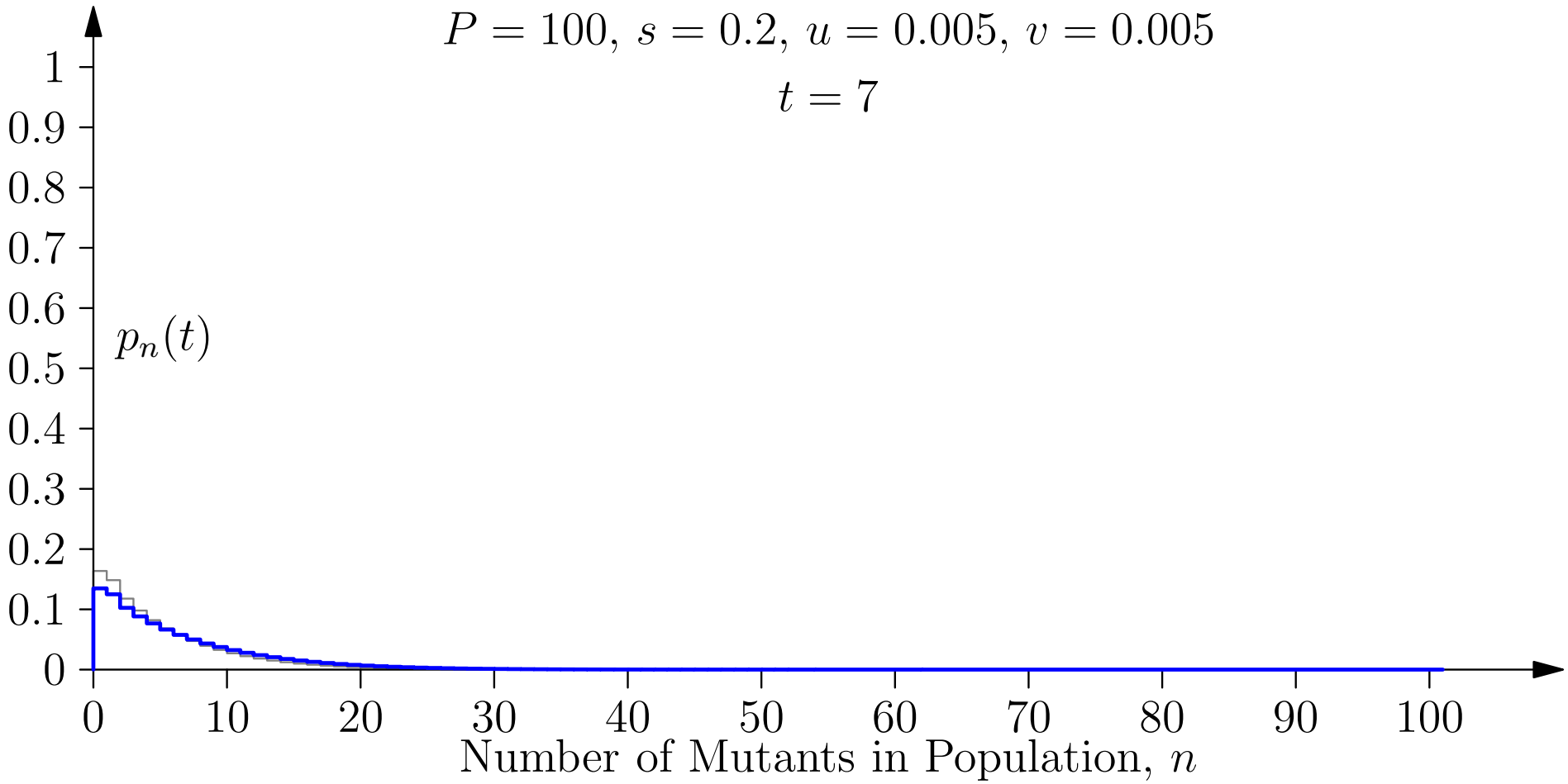

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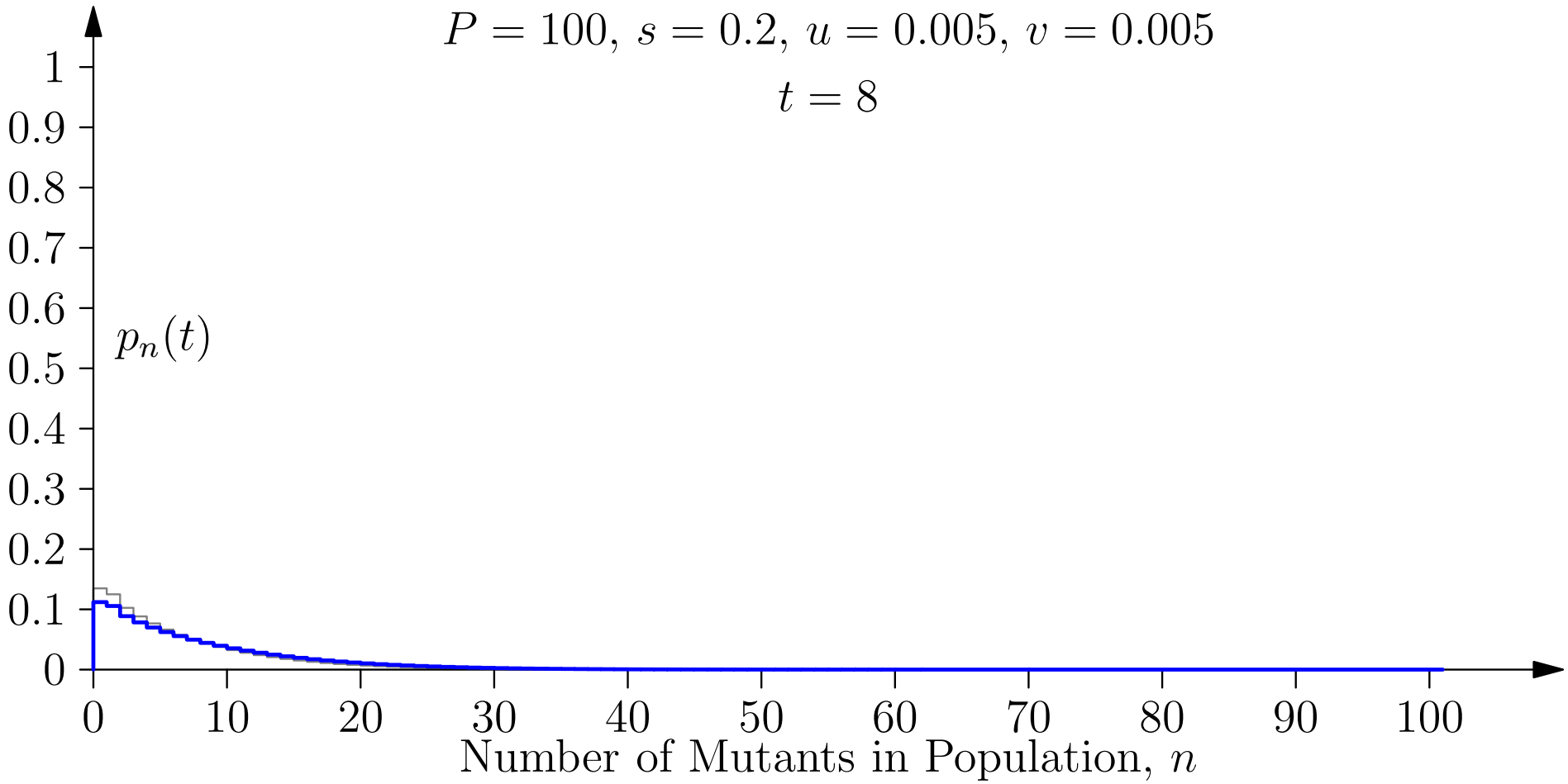

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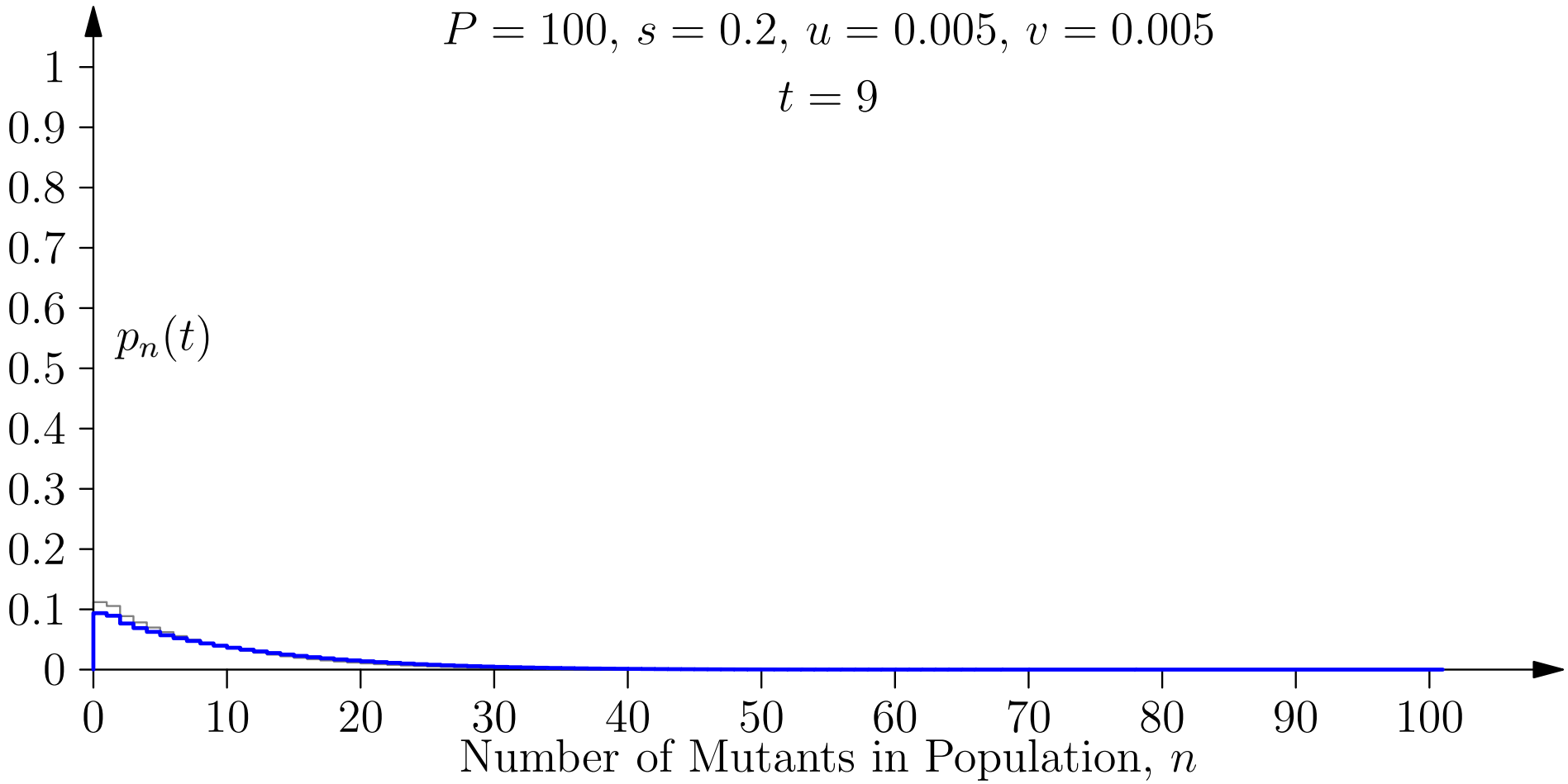

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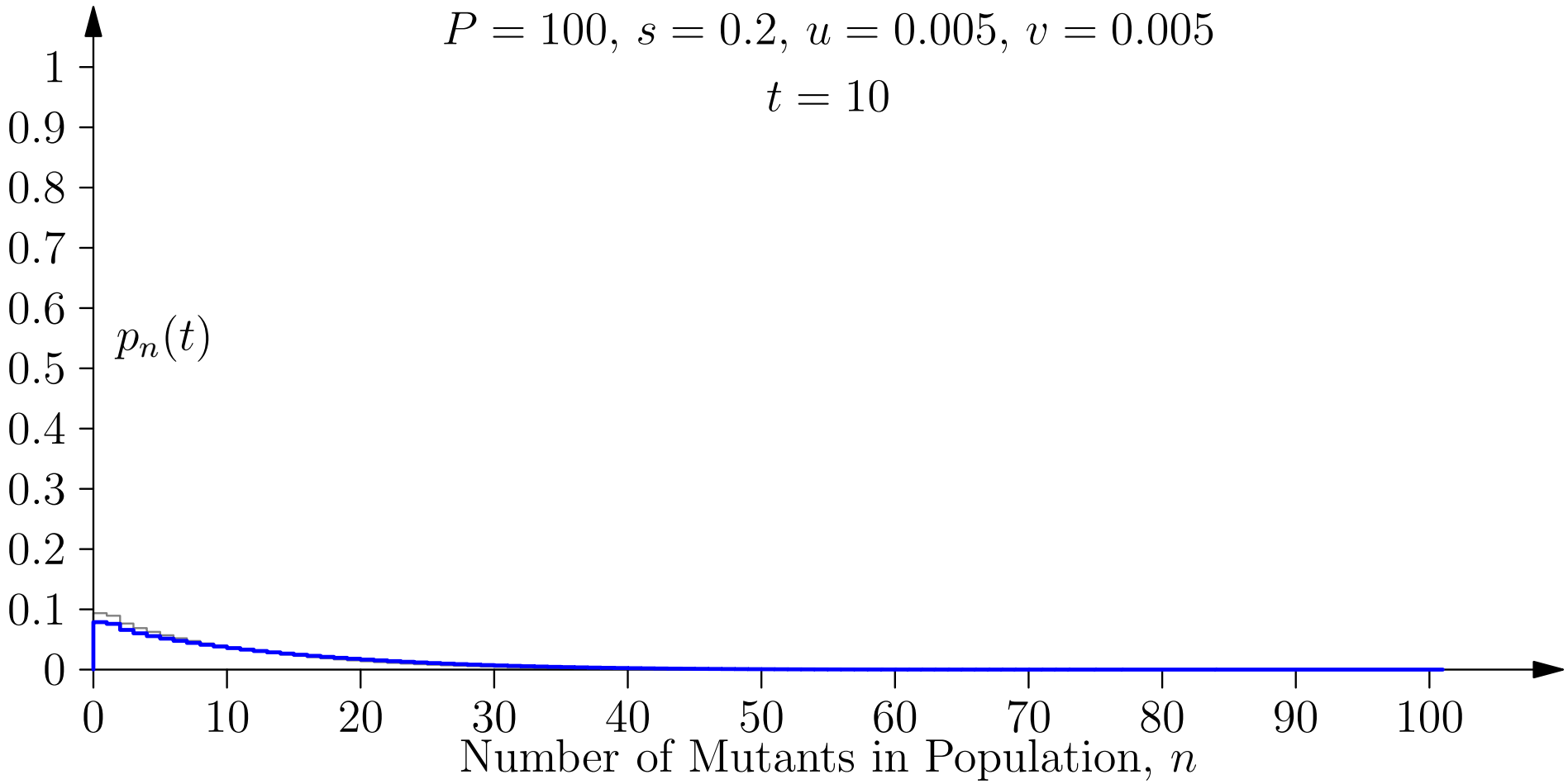

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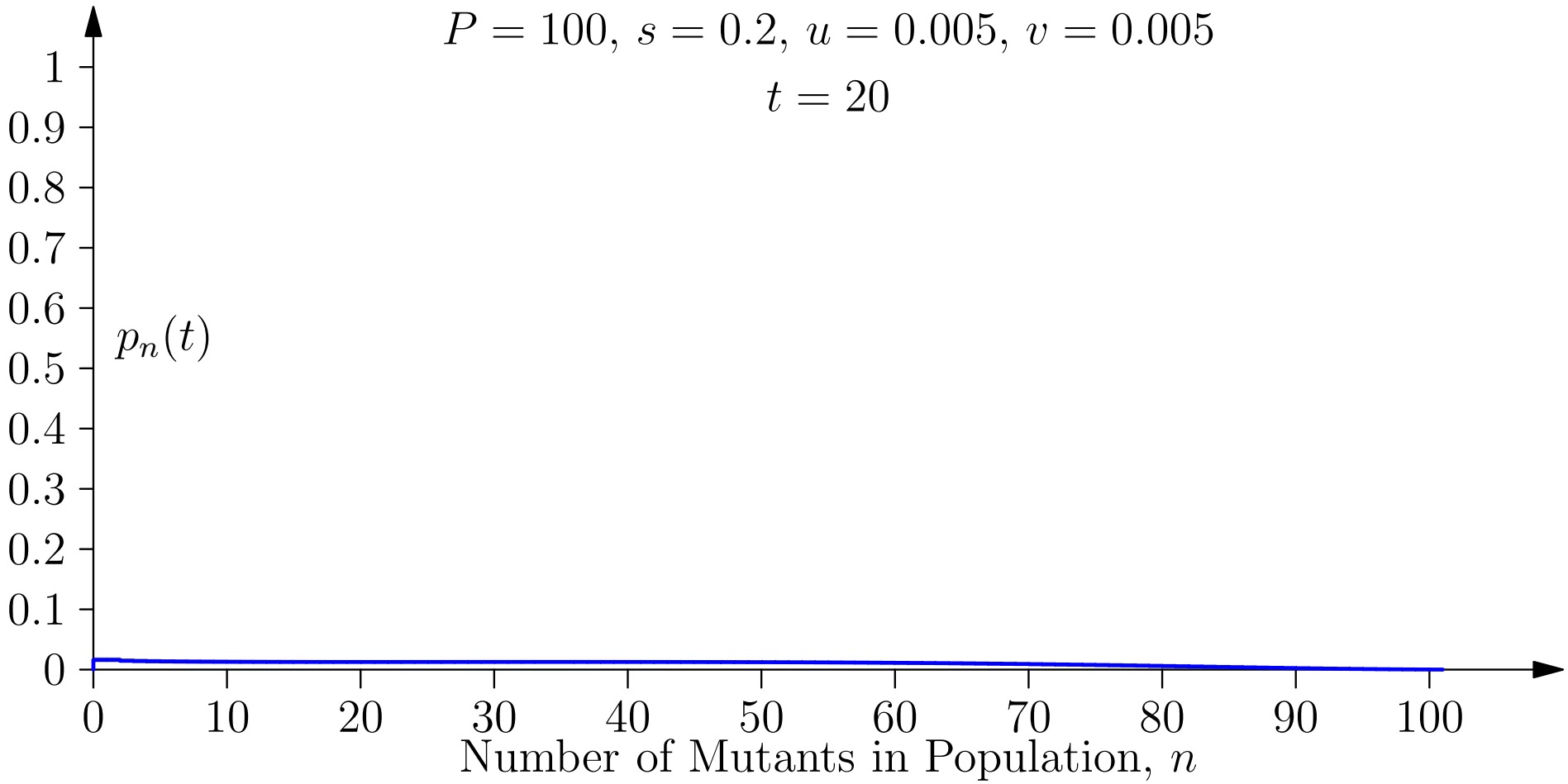

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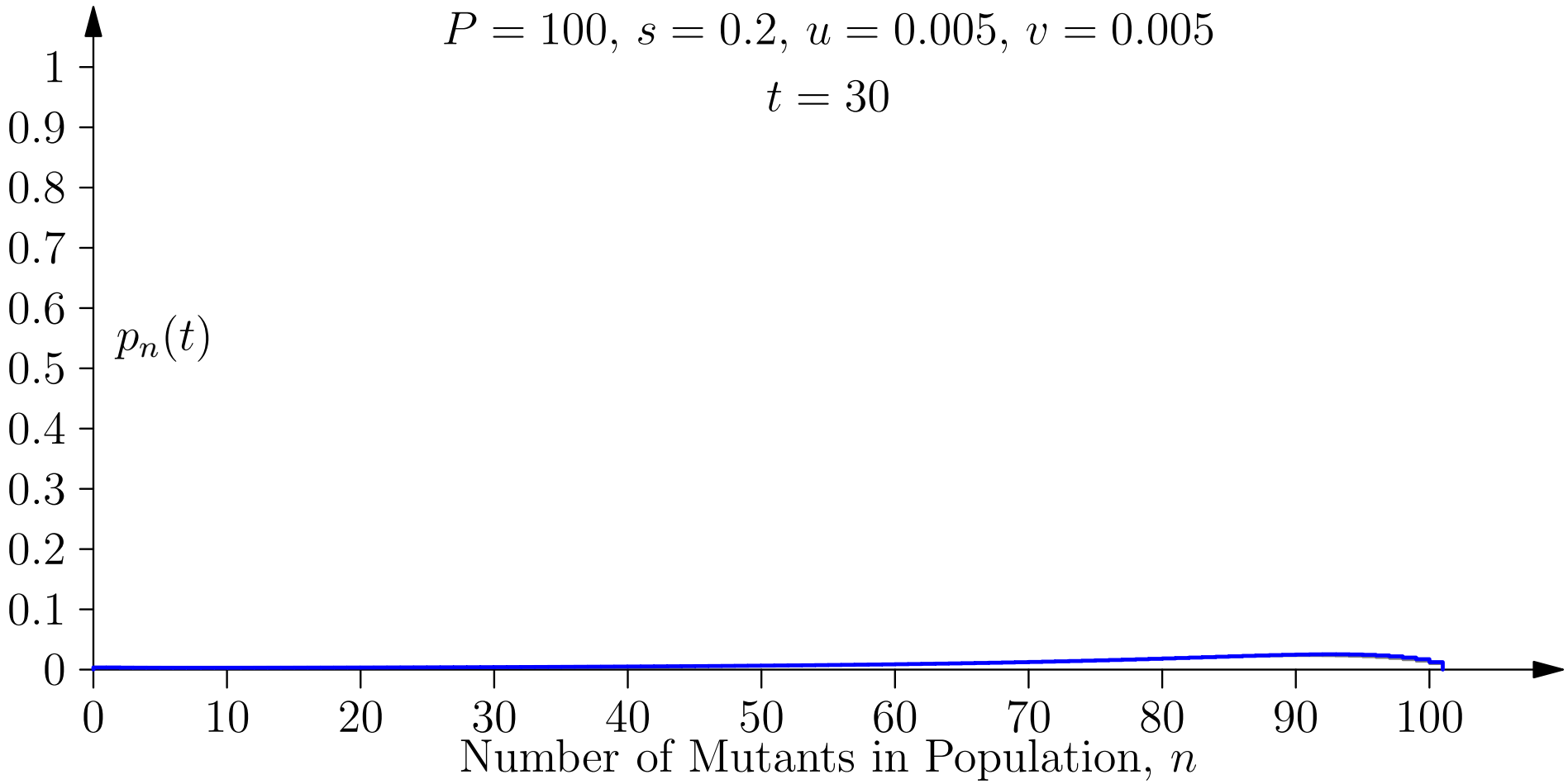

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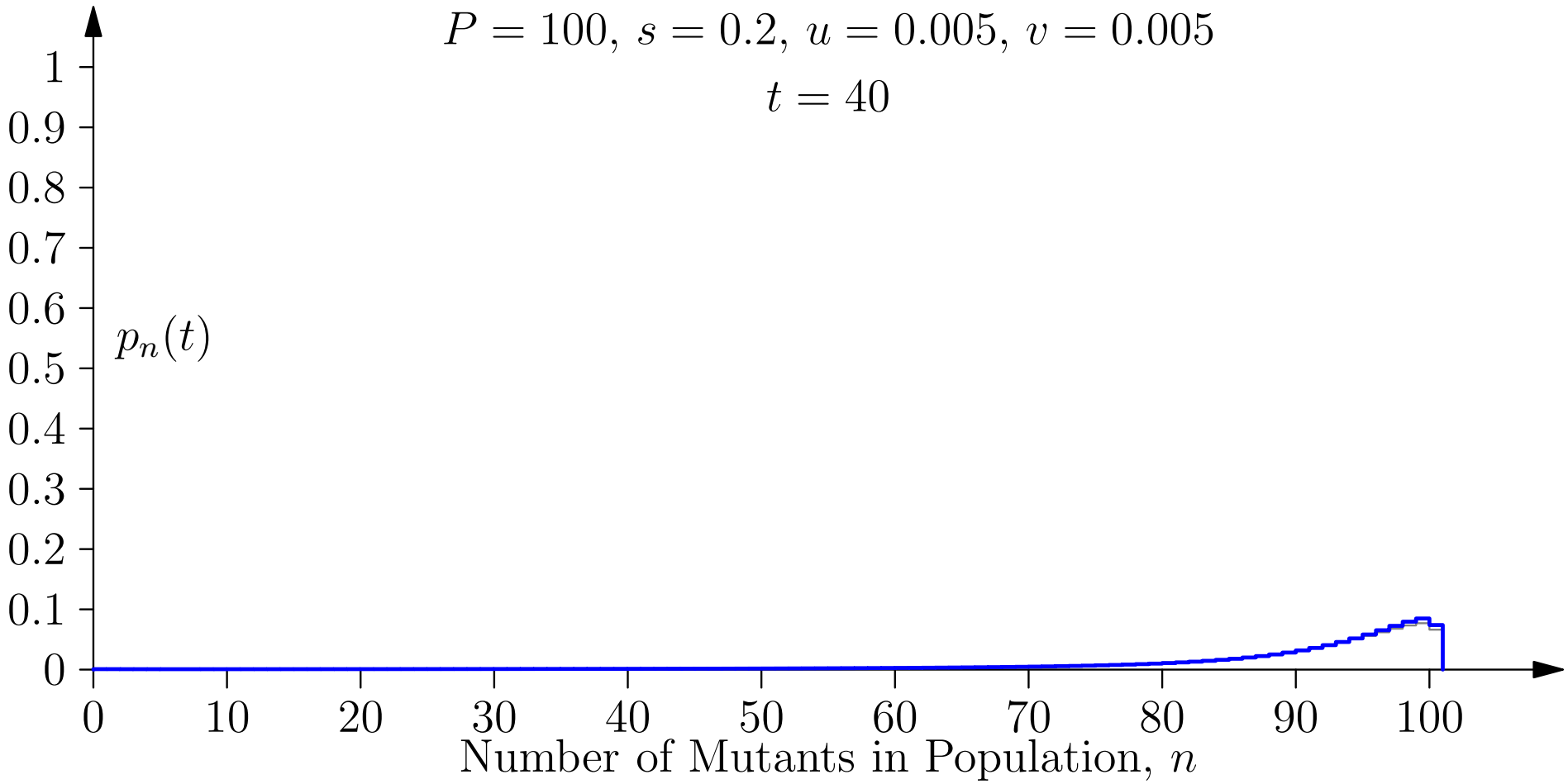

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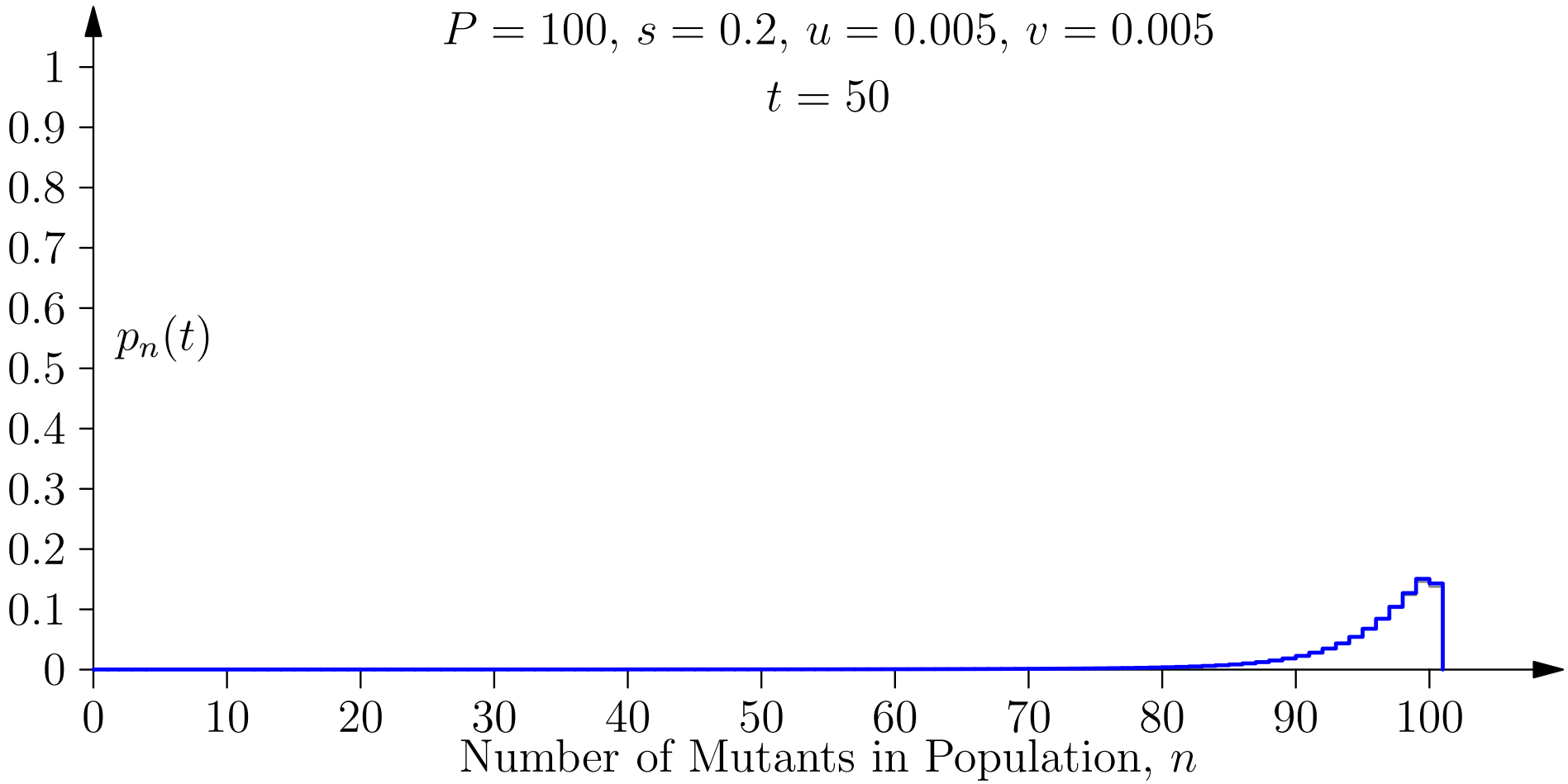

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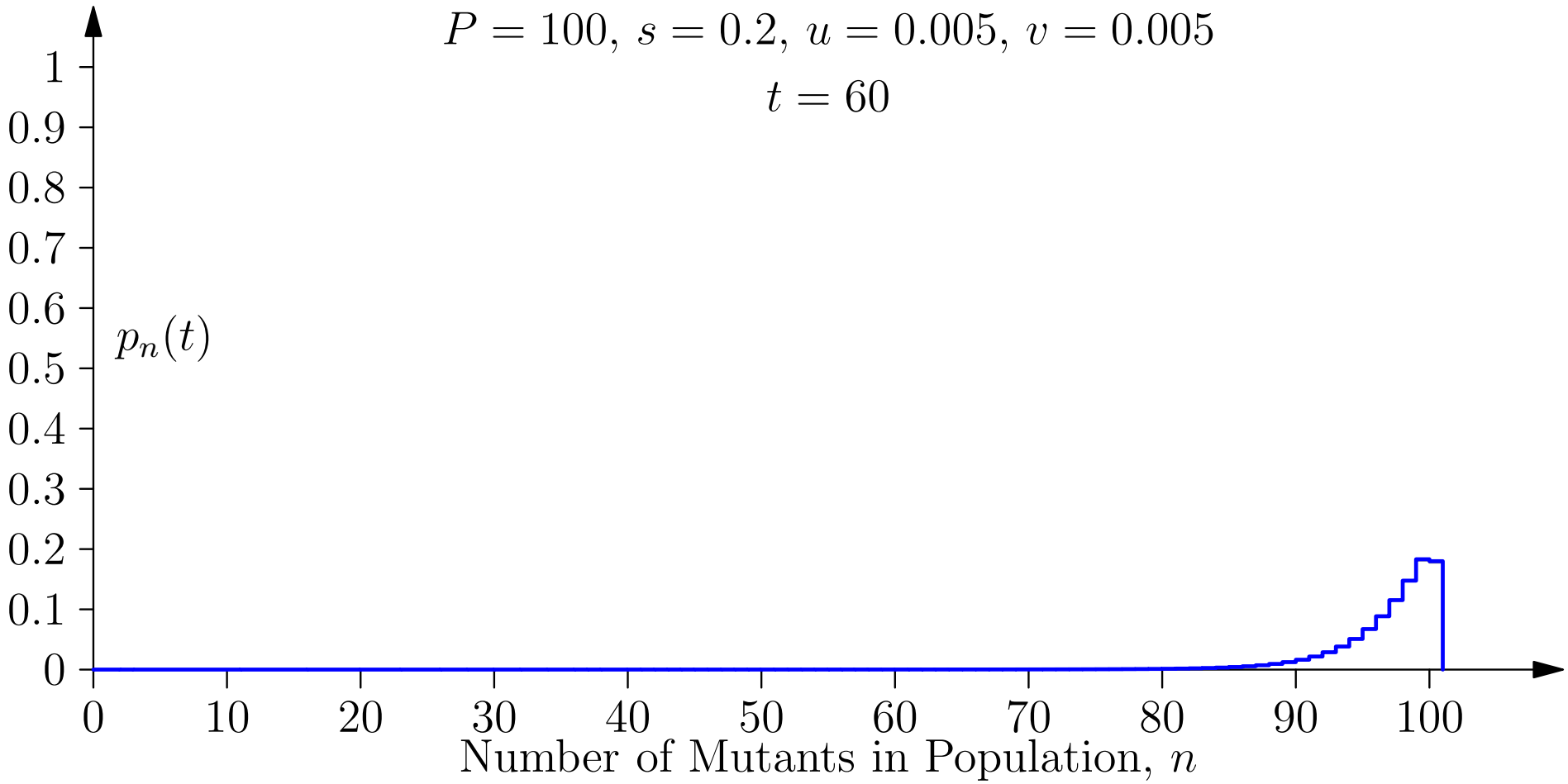

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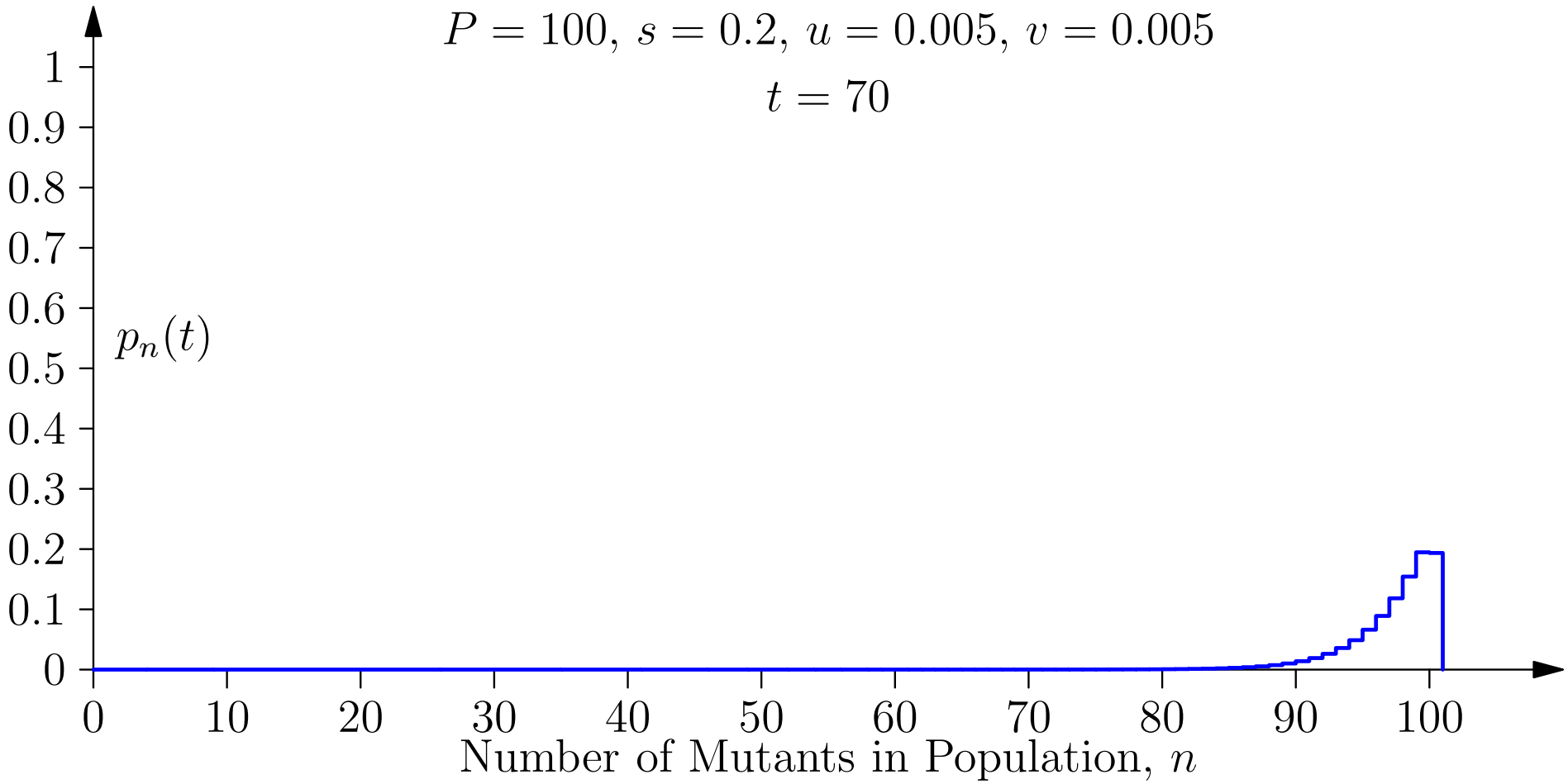

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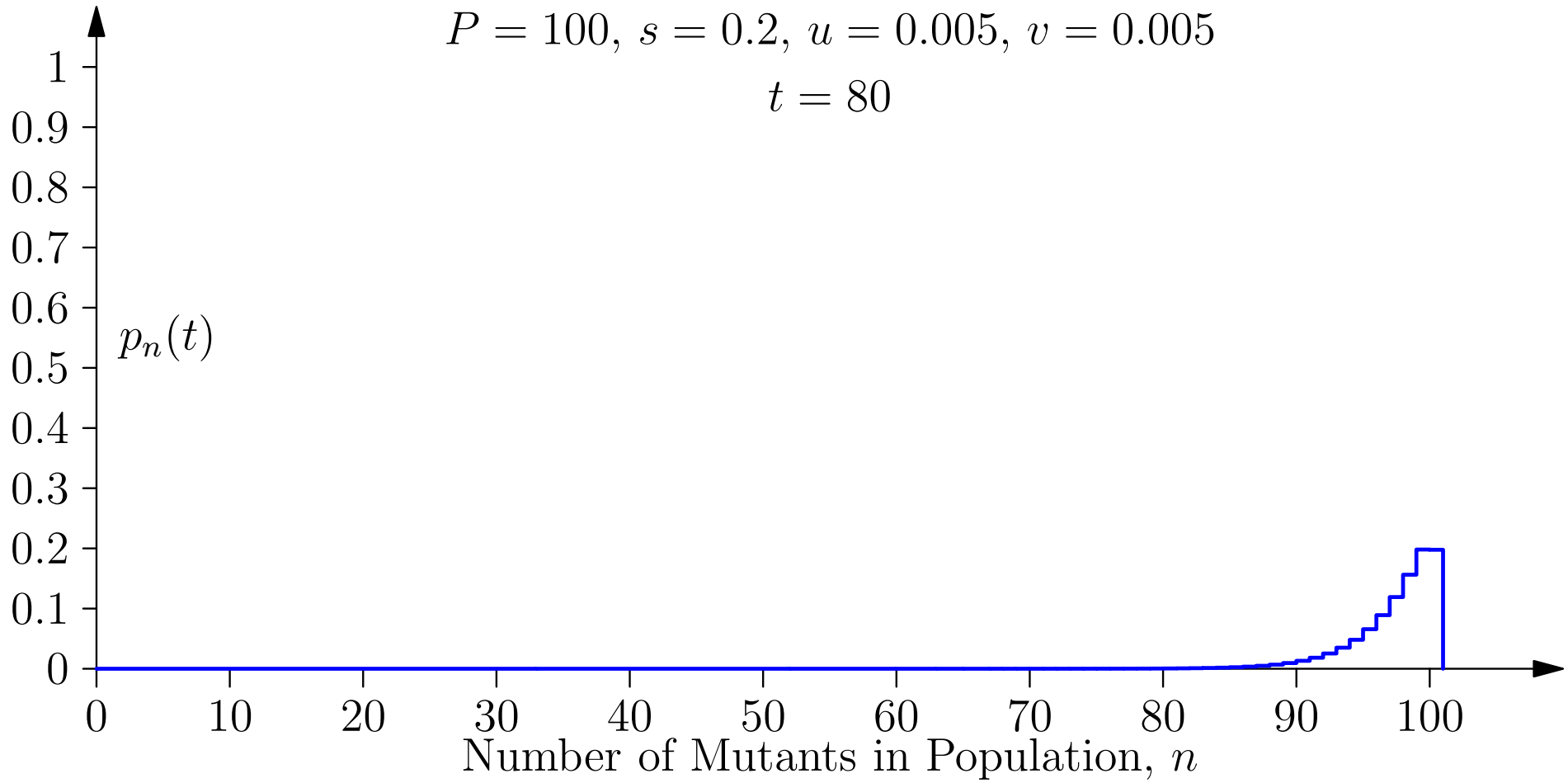

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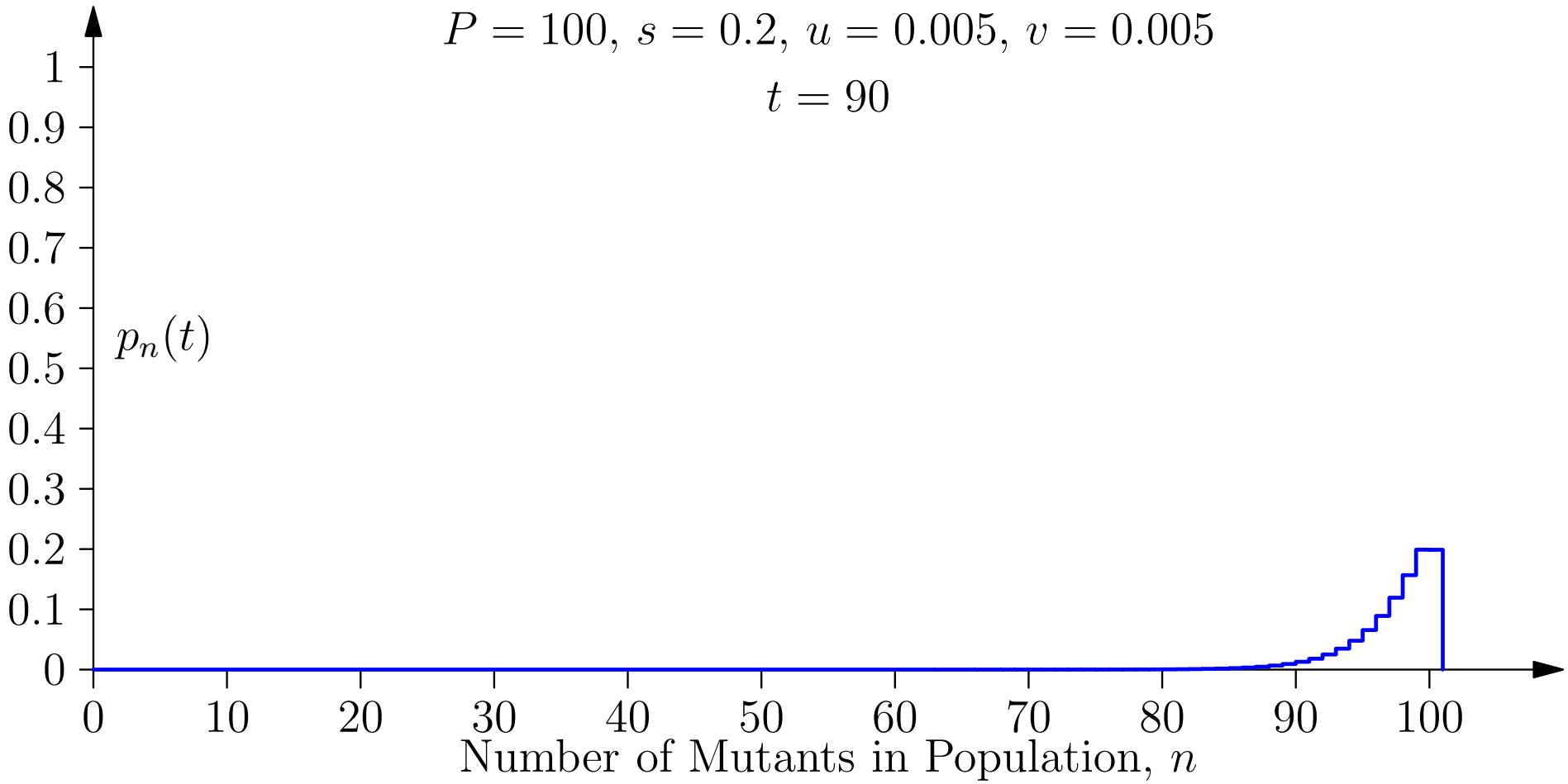

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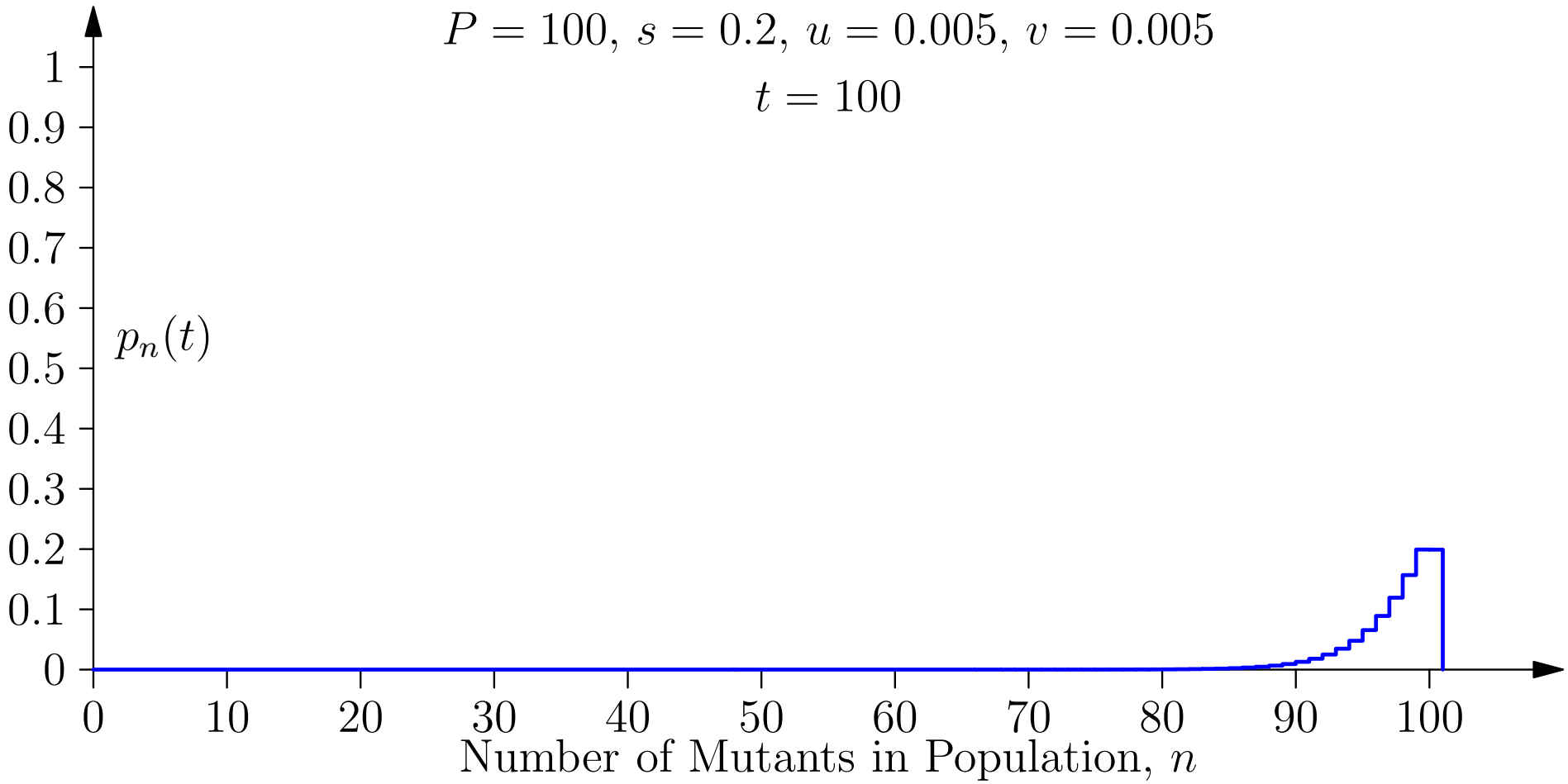

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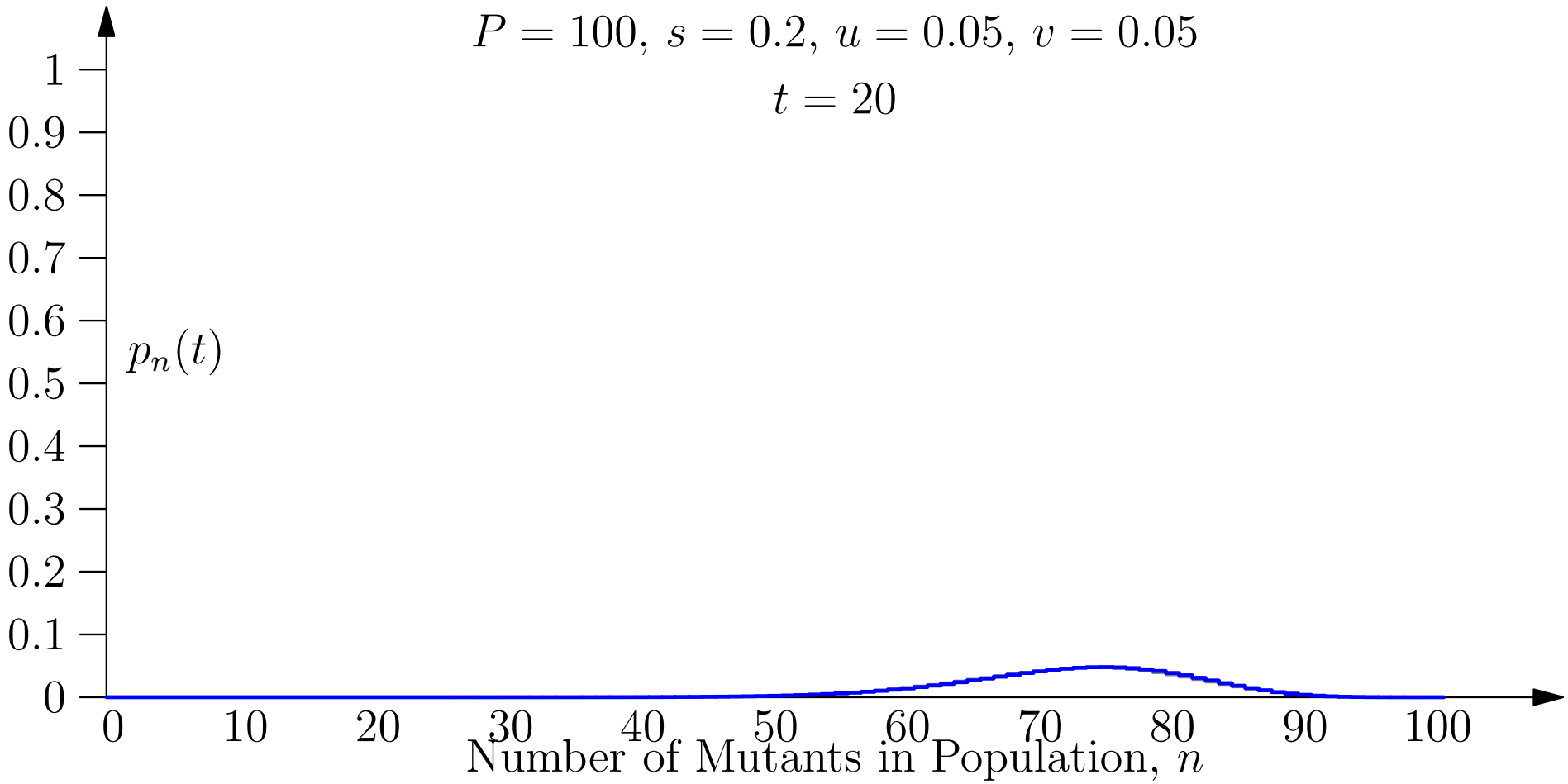

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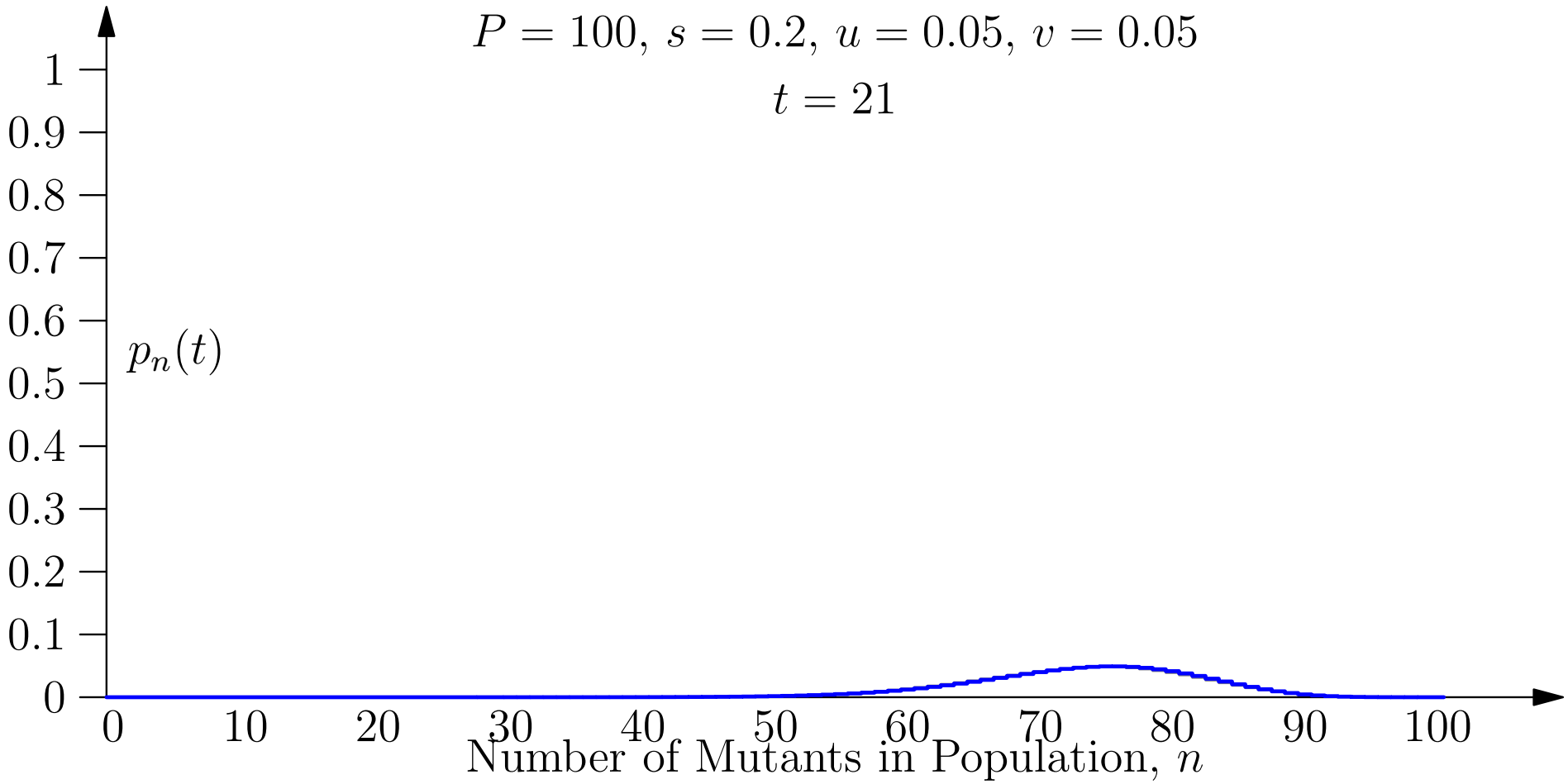

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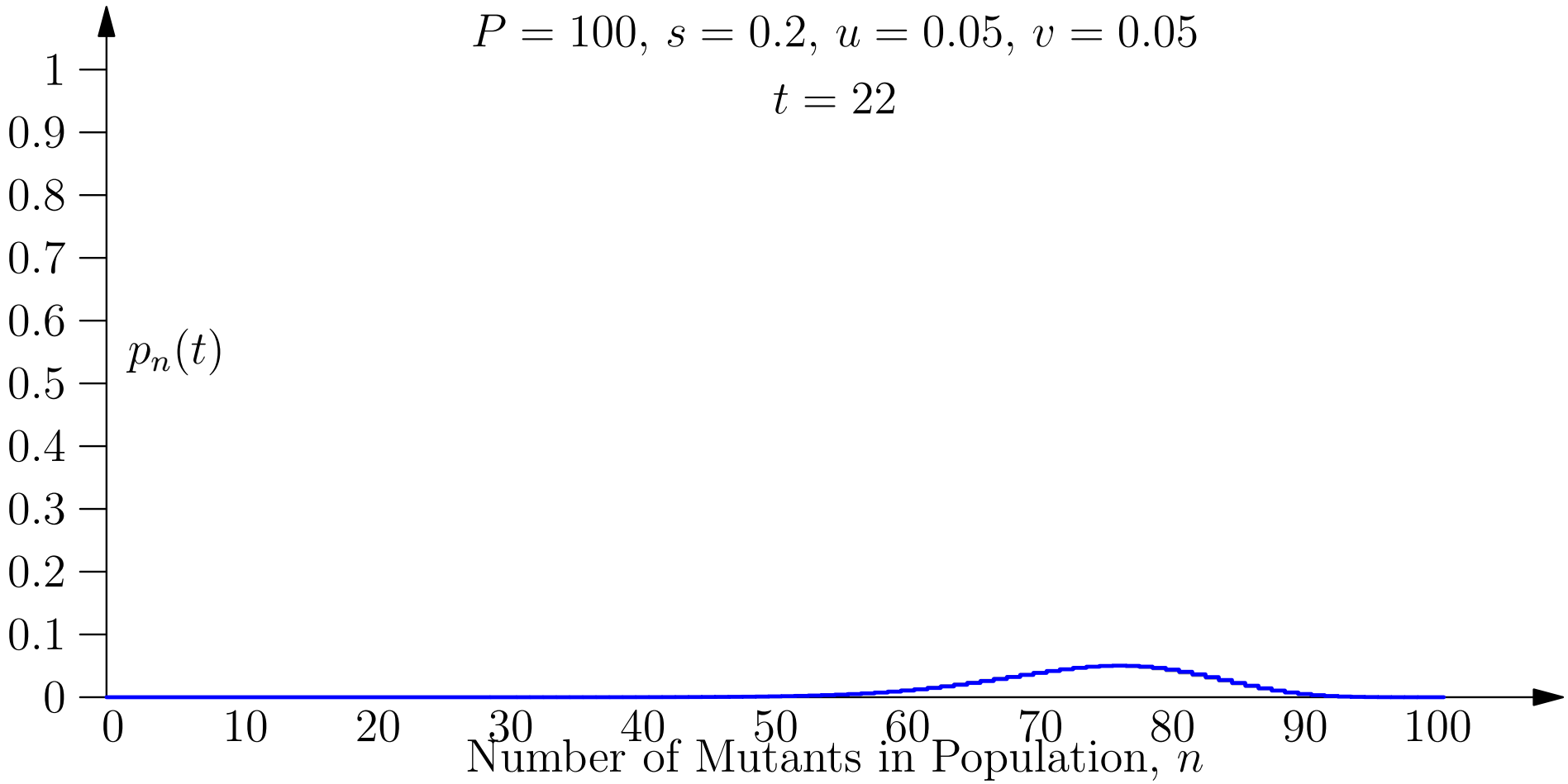

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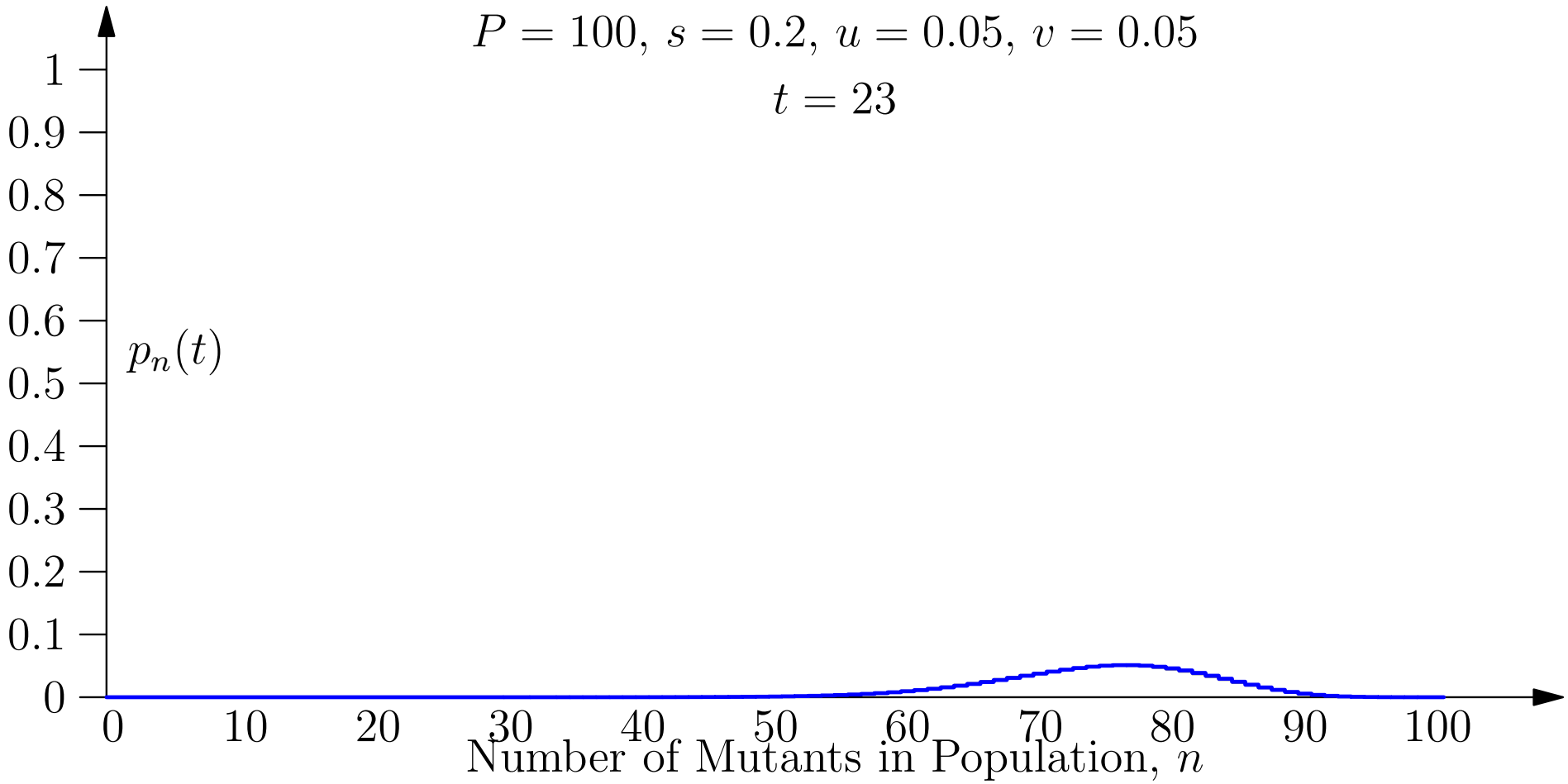

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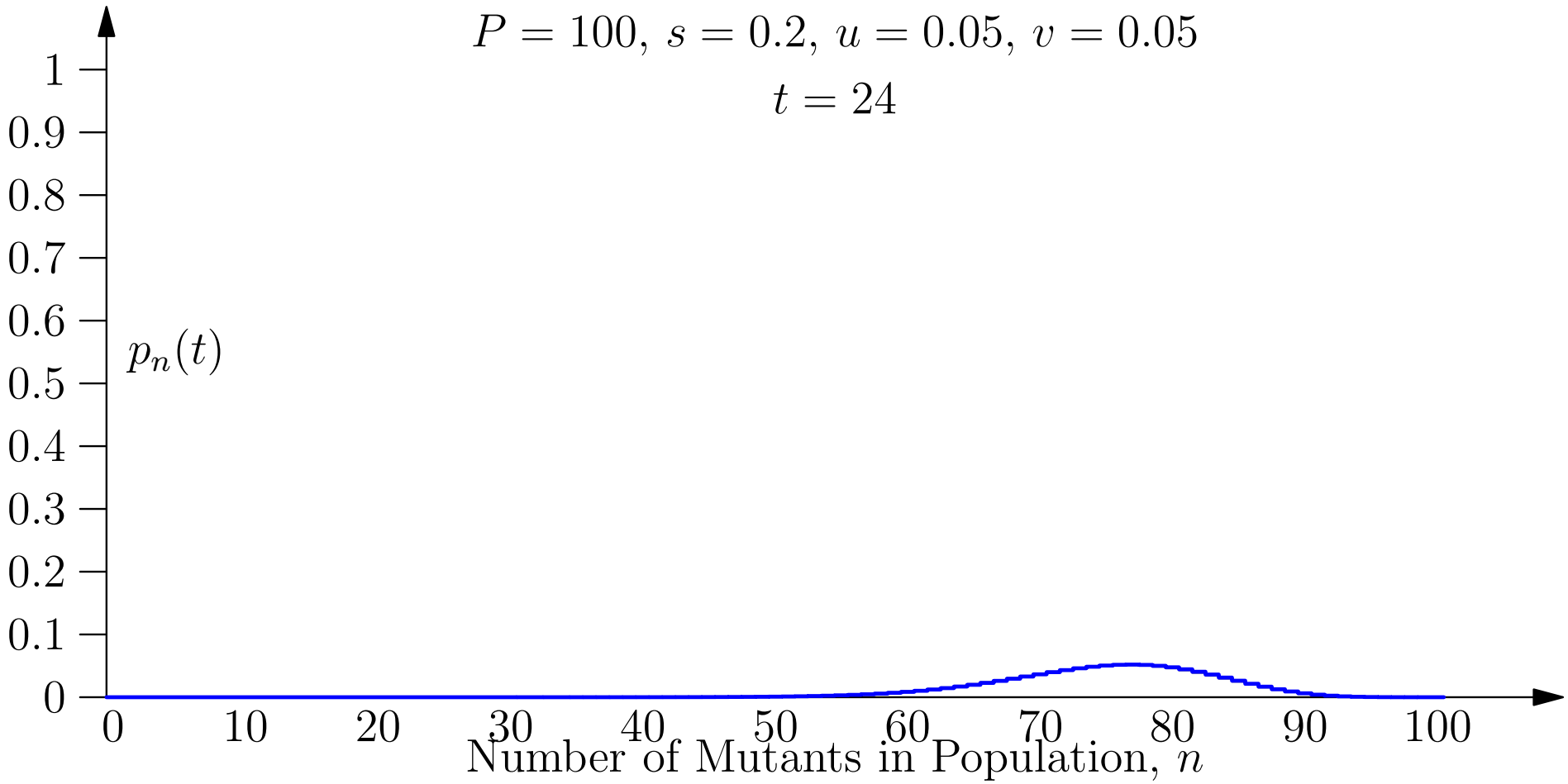

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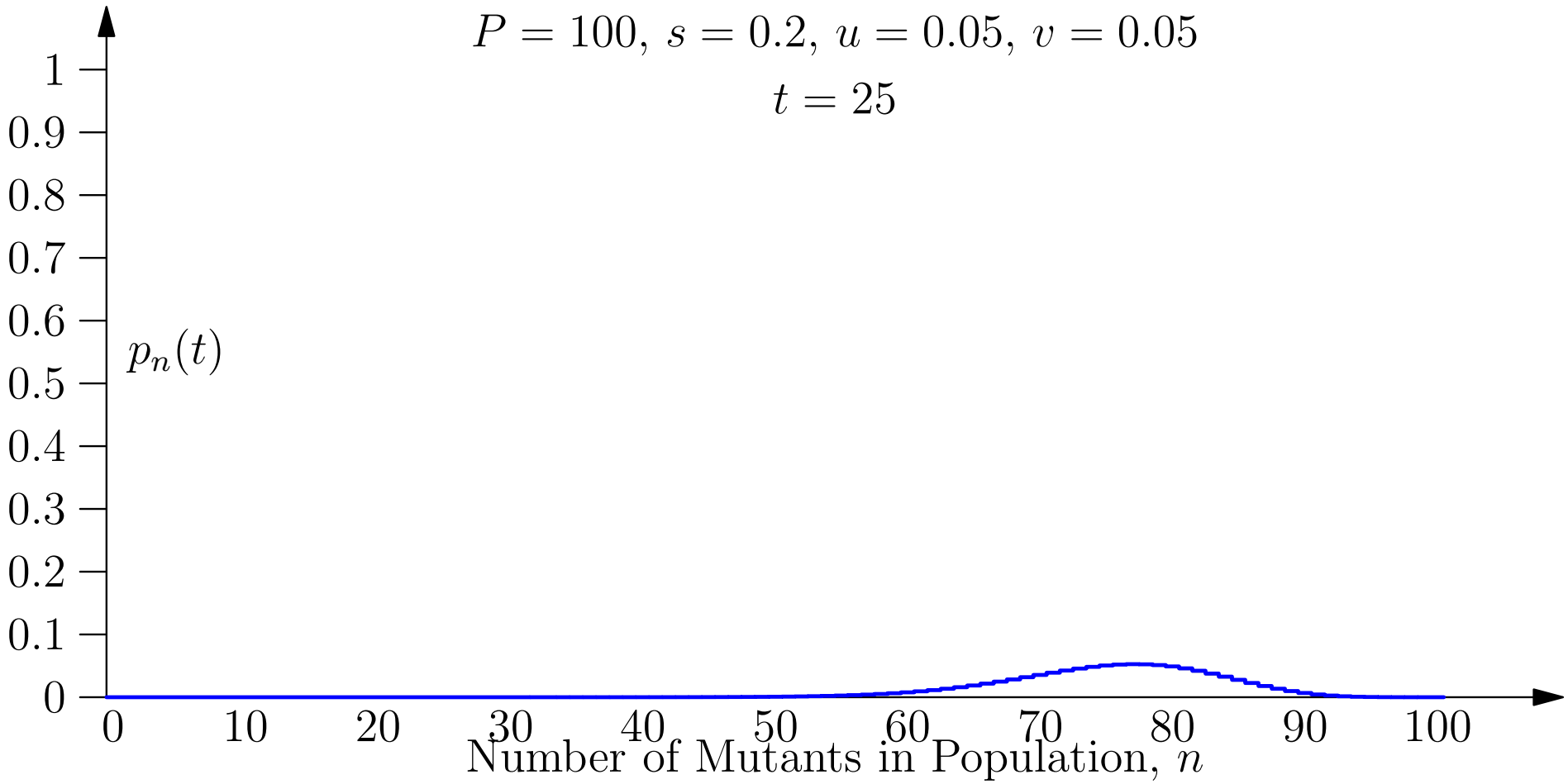
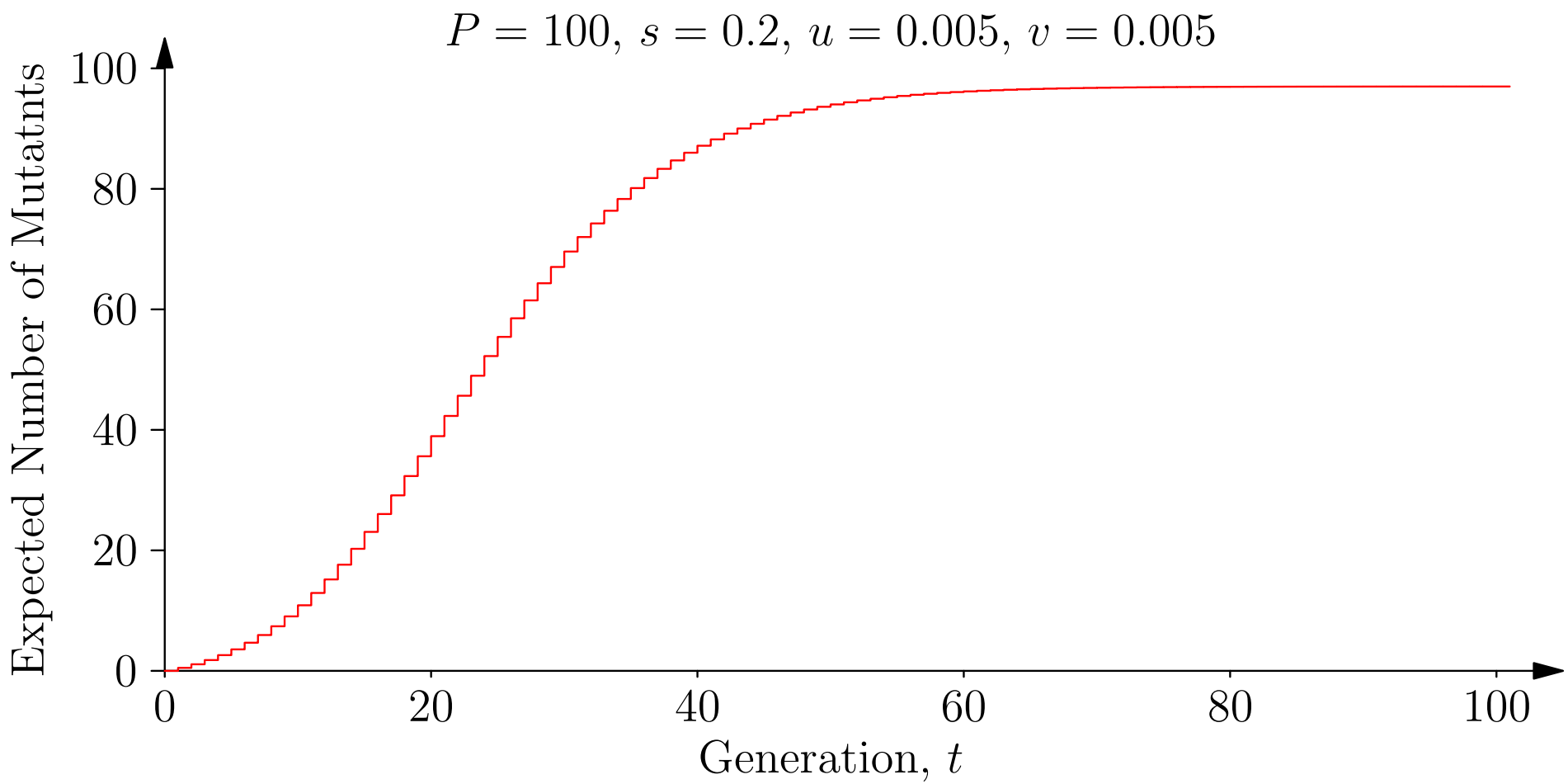


Mean Number of Mutants



Problems with Markov Chain Analysis

- The Markov Chain analysis is exact and provides a full picture of the probability of the population being in a certain state
- Unfortunately, the transition matrix $W(n \to n')$ is of size the number of states squared
- The number of states for a population of size P is $P + 1$
- For a large population writing the transition matrix becomes infeasible
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